How fiber orientation steers in-plane elastic waves: energy channeling, its limits, and manufacturable designs

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Abstract

Question. How much control over an in-plane elastic wave field can the fiber orientation of a continuous-fiber composite exert, at fixed material and mass and under the tow-curvature limit of automated placement? By what mechanism does it act, and what bounds it?
Hypothesis. In a strongly anisotropic lamina, wave energy is skewed toward the fiber direction: by up to 65° for CFRP, so that 88% of quasi-P wave normals carry their energy within 20° of the fibers. A fiber field should therefore act as a designable field of energy streamlines. Its robust effect should be *redirection* of energy, which is broadband. It should vanish for isotropic material, grow with anisotropy, and be limited by how sharply tows may turn.
Analysis. We solve the genuine two-dimensional orthotropic time-harmonic problem in `dolfinx`, using an exact real four-field split and automatic discrete adjoints, with a smooth radial-basis orientation field and a pointwise tow-curvature constraint. We decompose every focus gain exactly into redirection and field growth, sweep frequency on fixed designs, compute the energy-flux field, re-optimize along an anisotropy family, and trace the curvature front.
Results.

- An orientation-only lens's 78× single-frequency focus gain factorizes into 5.8× redirection (target share 1.47% → 8.6%) and a 13.4× resonant growth of the whole field. With a better absorber ($|R| \leq 0.13$ instead of 0.31–0.34) the re-optimized lens gives $70\times = 4.4\times$ redirection $\times\, 15.9\times$ growth: the split is robust, the redirection is absorber-dependent.
- Only the redirection persists off-design: it stays $\geq 3.7\times$ over $\omega \in [37.5, 60]$ ($\geq 3.6\times$ over $[40, 60]$ with the better absorber), while the gain collapses to 6.6× at −6%.
- The energy flux follows the local fibers: its mean deviation from them is 6.7° at the design frequency and at most 11.6° across the band tested; in curved fields that were never optimized it is only 1.0–4.2°.
- Redirection is exactly absent for isotropic material and grows with anisotropy. It is unaffected while the minimum tow radius is at most about 0.7 along-fiber wavelengths; beyond it the best design found declines gradually.
- A cloak reduces the far-field mismatch 36× at its design frequency (a value tuned to its mesh: 26× on a finer mesh, 40× when re-optimized there) but only 7.5–14.9× at ±5%, as expected for a full cloak that relies on interference.

Conclusion. Fiber orientation steers in-plane elastic energy by channeling it along the fibers. The transferable, broadband capability is redirection of a few times. Large single-frequency gains of finite plates are mostly resonance and should not be reported as steering. Designs should be optimized and judged over a band.

Keywords: elastic waves; fiber steering; anisotropic plates; energy channeling; topology optimization; cloaking; automated fiber placement.

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1 Introduction

Motivation. Automated fiber placement and continuous-fiber printing lay composite tows along curved paths, so the local fiber orientation $\theta(\boldsymbol{x})$ of a part becomes a design field. That field is routinely designed for *static* performance: stiffness and strength under a minimum turning-radius constraint, with the toolpath carried through the optimization [11]. A carbon/epoxy lamina is also strongly *acoustically* anisotropic. Its quasi-longitudinal wave speed differs by a factor of almost four between the fiber and transverse directions. This raises the possibility that the same manufacturing freedom, with no change of material or mass, could steer in-plane elastic waves: concentrate vibration where it is wanted, keep it from where it is not, or hide an obstacle from it.

1.1 Related work and positioning

Fiber-orientation design. Curvilinear fiber formats were introduced to raise buckling loads and relieve stress concentrations [1]. Variable-stiffness laminates are now designed routinely for stiffness and buckling [2]. Orientation is also optimized element-wise in discrete-material and continuous-orientation topology optimization [3, 4]. Continuous-fiber additive manufacturing [5] has made arbitrary in-plane fiber fields buildable. Wong et al. integrated toolpath generation and a local fiber-curvature limit into compliance-minimizing topology optimization [11]. We adopt their CS-RBF orientation field and curvature constraint. This work targets *static*, *modal* or forced-vibration objectives: the fundamental frequency [6], the natural modes of curvilinear-fiber laminates [7], and minimum sound radiation from vibrating laminates by discrete orientation selection [8]. Anisotropic composite layouts have also been designed to shield, focus and cloak *diffusive* heat flux [9, 10]. Olsson and Wall chose fiber directions for exact partial cloaking in the inextensible limit [16]; to our knowledge, fiber-orientation fields have not been *optimized*, under the manufacturing constraints of real tows, to steer elastic waves.

Wave control by graded media. Transformation optics [12] and transformation elastodynamics [13, 14, 21, 15] prescribe material fields that bend waves arbitrarily. The required media are singular at the inner surface of a cloak. With isotropic stiffness the density must become infinite there, and a finite-mass cloak needs anisotropic (pentamode) stiffness instead [14]; for in-plane elasticity the transform keeps the Navier form but with a tensor that is singular at the inner boundary and lacks the minor symmetries of a Cauchy solid [21, 15], and in general the transformed equations

are not of Cauchy-elastic form [13], so no lamina can realize them. Pre-stressing a hyperelastic solid is another route to the required properties [19, 20]. Olsson and Wall showed that inextensible fibers in a variable direction give form-invariant equations of motion and exact partial cloaking at all frequencies [16]; this is the idealized limit of unbounded anisotropy, which a real lamina does not reach. Ground-plane (carpet) cloaks, which only need to mimic a flat reflecting surface, can be broadband with bounded (natural or layered) anisotropy [17, 18]. In practice, elastic lenses and wave guides use graded index fields made from thickness variations [22] or phononic crystals [23], mostly for flexural waves in isotropic plates. Adjoint inverse design of plate wave devices has used pinning, mass loading and grading [24]. Topology optimization of wave problems is well developed for band gaps, acoustics and photonics [25, 26, 30], including low-contrast optical cloaks [27] and acoustic cloaks [28], and the single-frequency optima it produces are known to be narrowband, which motivates band-averaged objectives [29, 28]. In these approaches the design variable is the *amount* of material or its local stiffness, not its *orientation*.

Waves in anisotropic plates. In anisotropic solids, energy does not travel along the wave normal: the group velocity is skewed toward stiff directions [31, 32]. In fiber-reinforced composite plates this focuses guided waves into the directions of minimum slowness, which are the stiff directions [33]. In some anisotropic plates the power flux of slow guided waves can even be perpendicular to the wave vector [34]. In non-destructive evaluation the skew is a nuisance to be compensated. Here it is the mechanism to be designed.

What is new here. We pose wave control with the fiber orientation as the *only* design field: material, thickness and mass are fixed, and the field is manufacturable under the tow-curvature limit of [11]. We identify the mechanism, energy channeling by group-velocity skew, and test it prediction by prediction. We also show that the large single-frequency gains such optimizations produce in finite plates are mostly resonant field build-up: an elementary exact identity separates that build-up from the steering itself, and, following band-objective practice in wave topology optimization [29, 28], we evaluate every design across a band.

1.2 Question, hypothesis and predictions

Scientific question.

> *How much control over an in-plane elastic wave field can the fiber orientation of a continuous-fiber composite exert, at fixed material and mass and under the tow-curvature limit of automated placement? By what mechanism does that control act, and what bounds it?*

Hypothesis. (Fig. 1.) In a strongly anisotropic lamina, the energy of a plane wave does not travel along its wave normal. It is skewed toward the fiber direction, by up to 65° for CFRP (Sec. 3). Most wave normals therefore carry their energy within a narrow cone about the fibers. We hypothesize that a fiber field acts as a field of *energy streamlines*: designing $\theta(\boldsymbol{x})$ designs where vibration energy goes, because energy is channeled along the fibers. Section 3 makes this precise. It predicts:

- **W1** An isotropic plate offers no orientation control at all.
- **W2** The robust effect of an optimized field is *redirection*: a larger share of the wave energy in the target. Because the energy skew of a homogeneous lamina does not depend on frequency, redirection is broadband. Any gain beyond redirection that appears at the design frequency is field build-up (resonance) and is narrowband.

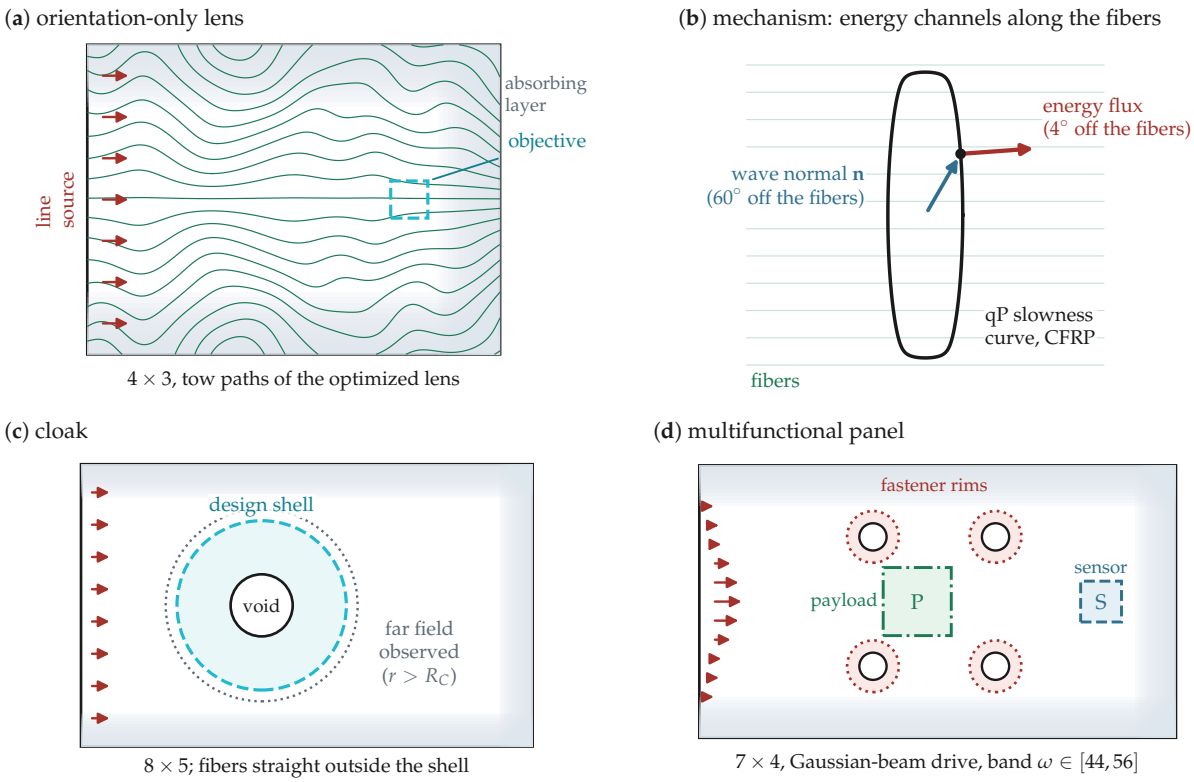


Figure 1: The three problems and the mechanism. **(a)** The lens: a 4×3 CFRP plate driven by an x-directed line source at its left edge, with absorbing layers on the other three edges (shaded), the tow paths of the optimized lens, and the objective region (dashed). **(b)** Why fibers channel energy: the quasi-P slowness curve of the CFRP lamina (fibers along x). The energy of a plane wave travels along the outward normal to this curve, so a wave whose normal is 60° off the fibers carries its energy only a few degrees off them (Sec. 3). **(c)** The cloak: a traction-free void cut from the mesh, the design shell in which the orientation may change (dashed), and the far-field observation region beyond R_C (dotted). **(d)** The multifunctional panel: four fastener holes with their rim regions (dotted), the payload P and sensor S, and a Gaussian-beam drive.

W3 In the optimized designs, the local energy flux follows the fiber direction.

W4 Redirection grows with the lamina anisotropy, tracking the share of wave normals that the lamina channels toward its fibers.

W5 Energy can follow a curving fiber field only as fast as the field may turn, so the tow-curvature limit bounds the redirection. Canceling the field of a finite scatterer in free space (a full cloak, not a ground-plane cloak) needs the singular anisotropy of transformation media. With the bounded anisotropy of a real lamina it must rely on interference, and we predict it to be narrowband.

1.3 Approach and findings

Approach. We solve the genuine two-dimensional orthotropic time-harmonic problem: the rotated plane-stress stiffness $\boldsymbol{D}_f(\theta)$ with $E_{f1}/E_{f2} = 14.6$, light structural damping, and an absorbing

sponge. A real four-field split runs it on a real-arithmetic `PETSc/dolfinx` build, and automatic discrete adjoints give the orientation sensitivity. The field is parametrized by smooth compactly supported radial basis functions and bounded by the tow curvature. We optimize a lens and a cloak, then test each prediction by decomposing the gains, sweeping frequency, measuring the energy-flux field, varying the anisotropy, and tightening the curvature limit. A band-averaged multifunctional panel shows how the results translate into a design practice.

Findings in brief.

- **W1.** Orientation is exactly inert for an isotropic lamina (a consistency check: this is an identity of the model, verified on the discrete gradient).
- **W2.** The lens's single-frequency target-energy gain of 78× factorizes exactly into a redirection of 5.8× (target share 1.47% → 8.6%) and a 13.4× growth of the whole interior field. Only the redirection survives a frequency change: it stays $\geq 3.7\times$ over $\omega \in [37.5, 60]$, while the gain falls to 6.6× and 10.1× at ±6%. With a better absorber the same split holds ($70\times = 4.4\times$ (G_φ) $\times$ 15.9× (growth)) at a lower redirection.
- **W3.** At the design frequency, the energy flux in the lens deviates from the local fiber direction by 6.7° on average; in curved fiber fields that were never optimized it deviates by only 1.0–4.2°, so channeling is a property of the lamina, not of the design.
- **W4.** Redirection grows with anisotropy.
- **W5, tow curvature.** Redirection is unaffected while the minimum tow radius is at most ≈ 0.7 along-fiber wavelengths; beyond it the best design found declines gradually (to about half by 1.2 wavelengths).
- **W5, cloak.** The cloak's 36× far-field reduction is confined to the design frequency (7.5× and 14.9× at ±5%), as predicted.
- **Consequence for design.** Single-frequency gains of finite plates overstate what orientation does. Redirection, band-averaged, is the meaningful measure.

2 Governing model and the anisotropy lever

Consider in-plane time-harmonic motion $\mathrm{Re}\{\mathbf{u}(\mathbf{x})e^{+\mathrm{i}\omega t}\}$ (with this sign convention, structural damping $(1+\mathrm{i}\eta)$ and a viscous sponge $+\mathrm{i}\omega c$ are both dissipative), $\mathbf{u} = (u_x, u_y)$, of a thin composite plate $\Omega \subset \mathbb{R}^2$ under plane-stress. With the engineering Voigt strain

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \left[\, \partial_x u_x,\ \ \partial_y u_y,\ \ \partial_y u_x + \partial_x u_y \,\right]^{\mathsf{T}}, \tag{1}$$

the fiber composite has the rotated orthotropic plane-stress stiffness

$$\mathbf{D}_f(\theta) = \mathbf{T}(\theta)\,\mathbf{Q}_f\,\mathbf{T}(\theta)^{\mathsf{T}}, \qquad \mathbf{T}(\theta) = \begin{bmatrix} c^2 & s^2 & -2sc \\ s^2 & c^2 & 2sc \\ sc & -sc & c^2 - s^2 \end{bmatrix}, \ \ (c,s) = (\cos\theta, \sin\theta), \tag{2}$$

where $\mathbf{Q}_f$ is the fiber stiffness in material axes,

$$\mathbf{Q}_f = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix}, \quad Q_{11} = \tfrac{E_{f1}}{1-\nu_{12}\nu_{21}},\ \ Q_{22} = \tfrac{E_{f2}}{1-\nu_{12}\nu_{21}},\ \ Q_{12} = \tfrac{\nu_{12}E_{f2}}{1-\nu_{12}\nu_{21}},\ \ Q_{66} = G_{12}, \tag{3}$$

with $\nu_{21} = \nu_{12}E_{f2}/E_{f1}$. We use a carbon/epoxy lamina $E_{f1} = 131$, $E_{f2} = 9$, $G_{12} = 5$, $\nu_{12} = 0.27$ (a modulus contrast $E_{f1}/E_{f2} \approx 14.6$), density $\rho = 1.6$. The time-harmonic displacement obeys

$$-\nabla\cdot\big[(1+\mathrm{i}\eta)\,\boldsymbol{\sigma}(\mathbf{u};\theta)\big] - \omega^2\rho\,\mathbf{u} + \mathrm{i}\,\omega\,c\,\mathbf{u} = \mathbf{f}, \qquad \boldsymbol{\sigma} = \mathbf{D}_f(\theta)\,\boldsymbol{\varepsilon}(\mathbf{u}), \tag{4}$$

where η is a small structural (hysteretic) damping that keeps the response off sharp undamped resonances (we use $\eta = 0.02$ in every computation, and test 0.01 and 0.04 in Appendix F; without it the field is speckle), and $c(\mathbf{x}) \geq 0$ is a mass-proportional absorbing sponge (§4.3). The orientation $\theta(\mathbf{x})$ is the sole design field.

Remark 1 (Isotropy annihilates the orientation lever)**.** *The stiffness of an* isotropic *material is a rotationally invariant tensor: if* $\mathbf{Q}_f$ *is isotropic (*$Q_{11} = Q_{22}$ *and* $Q_{66} = \frac{1}{2}(Q_{11} - Q_{12})$*) then* $\mathbf{T}(\theta)\mathbf{Q}_f\mathbf{T}(\theta)^{\mathsf{T}} = \mathbf{Q}_f$ *for every* θ*, so* $\mathbf{D}_f(\theta)$ *— and hence the entire operator* (4) *— is* independent of θ*. Consequently* $\partial\mathbf{D}_f/\partial\theta \equiv \mathbf{0}$*, every orientation sensitivity below (which we show is built from* $\partial\mathbf{D}_f/\partial\theta$*) vanishes identically, and fiber orientation is a* dead design variable *for an isotropic plate. Fiber anisotropy is therefore* necessary *for orientation-based wave control. How much control it buys is the subject of Sec. 3 and of the anisotropy study of Sec. 6.3.*

3 Theory: how a fiber field steers wave energy

3.1 Plane waves in a rotated lamina

Consider a homogeneous lamina with fiber angle θ and a plane wave $\mathbf{u} = \boldsymbol{a}\,e^{\mathrm{i}(\omega t - k\,\mathbf{n}\cdot\mathbf{x})}$ with unit wave normal $\mathbf{n} = (\cos\phi, \sin\phi)$. The engineering strain is $\boldsymbol{\varepsilon} = -\mathrm{i}k\,\mathbf{N}(\mathbf{n})\boldsymbol{a}$, with $\mathbf{N}(\mathbf{n}) = \left[\begin{smallmatrix} n_1 & 0 \\ 0 & n_2 \\ n_2 & n_1 \end{smallmatrix}\right]$. Substituting into (4) without damping gives the Christoffel problem

$$\boldsymbol{\Gamma}(\mathbf{n};\theta)\,\boldsymbol{a} = \rho c^2 \boldsymbol{a}, \qquad \boldsymbol{\Gamma}(\mathbf{n};\theta) = \mathbf{N}(\mathbf{n})^{\mathsf{T}}\mathbf{D}_f(\theta)\,\mathbf{N}(\mathbf{n}), \qquad c = \omega/k. \tag{5}$$

Its two eigenvalues give the phase speeds of the quasi-longitudinal (qP) and quasi-shear (qS) waves.

Proposition 1 (Only the relative angle matters)**.** $\boldsymbol{\Gamma}(\mathbf{n};\theta)$ *is unitarily similar to* $\boldsymbol{\Gamma}(\mathbf{n}_\psi;0)$*, where* $\mathbf{n}_\psi$ *is the unit vector at the relative angle* $\psi = \phi - \theta$*. Hence* $c = c(\psi)$*, and for an isotropic lamina* c *is independent of* ψ*.*

Proof. Let $\boldsymbol{R}$ be the in-plane rotation by θ. Then $\mathbf{N}(\boldsymbol{R}\mathbf{n}_\psi) = \mathbf{T}(\theta)^{-\mathsf{T}}\mathbf{N}(\mathbf{n}_\psi)\boldsymbol{R}^{\mathsf{T}}$ (strain rotation), and $\mathbf{D}_f(\theta) = \mathbf{T}\mathbf{Q}_f\mathbf{T}^{\mathsf{T}}$. Hence $\boldsymbol{\Gamma}(\mathbf{n};\theta) = \boldsymbol{R}\,\mathbf{N}(\mathbf{n}_\psi)^{\mathsf{T}}\mathbf{Q}_f\mathbf{N}(\mathbf{n}_\psi)\boldsymbol{R}^{\mathsf{T}} = \boldsymbol{R}\,\boldsymbol{\Gamma}(\mathbf{n}_\psi;0)\,\boldsymbol{R}^{\mathsf{T}}$. If $\mathbf{Q}_f$ is isotropic, then $\mathbf{D}_f(\theta) = \mathbf{Q}_f$ (Remark 1) and $\boldsymbol{\Gamma}$ depends on $\mathbf{n}$ only through its rotation. □

3.2 Energy velocity and skew

For a lossless homogeneous medium, the time-averaged energy flux of a plane wave equals its mean energy density times the group velocity $\boldsymbol{c}_g = \nabla_{\boldsymbol{k}}\omega$ [31]. Writing $\omega = k\,c(\psi)$ in polar wavevector coordinates,

$$\boldsymbol{c}_g = c(\psi)\,\mathbf{n} + c'(\psi)\,\mathbf{n}^{\perp}, \qquad \tan\delta(\psi) = \frac{c'(\psi)}{c(\psi)}, \tag{6}$$

so the energy travels at the angle $\psi + \delta(\psi)$ from the fiber. Here δ is the *skew* between the energy direction and the wave normal. For the CFRP lamina ($E_{f1}/E_{f2} = 14.6$) the qP speed falls from 9.07 along the fiber to 2.38 across it, and the skew reaches 65° at $\psi = 75°$.

- A qP wave whose normal is 60° off the fibers carries its energy only 4° off them.
- The qS skew reaches 59°.

Channeling. We call the fraction of wave normals whose energy direction lies within 20° of the fiber the *channeling share* s_{20} of the lamina. For an isotropic lamina it is 22%, the purely geometric fraction 20/90. Along the family $\mathbf{Q}(s)$, which keeps the isotropic invariants and scales the anisotropic ones (Sec. 6.3), it rises monotonically:

E_{f1}/E_{f2}	1.0	1.7	2.9	5.4	14.6	31.9
max qP skew	0°	18°	35°	50°	65°	71°
s_{20}	22%	42%	60%	75%	88%	94%

A CFRP lamina therefore sends the energy of almost every incident direction along its fibers. This is the physical content of the hypothesis. When θ varies slowly on the wavelength scale, ray (WKB) theory applies locally, and the energy paths of a wave field in a steered plate approximate the integral curves of the fiber field. The fiber field is then a designable field of energy streamlines.

3.3 Predictions

- **W1 (isotropy).** By Proposition 1 and Remark 1, an isotropic lamina makes the operator independent of θ. Every orientation sensitivity vanishes identically. This is an identity of the model, not an empirical prediction; we use it as a consistency check of the discrete gradient.
- **W2 (redirection is the robust effect).** Write the target-energy gain of a design against straight fibers as

$$G = \frac{E_f}{E_f^0} = \underbrace{\frac{E_f/E_{\text{int}}}{E_f^0/E_{\text{int}}^0}}_{G_\varphi \text{ (redirection)}} \times \underbrace{\frac{E_{\text{int}}}{E_{\text{int}}^0}}_{\text{field growth}}, \tag{7}$$

  where E_f is the energy in the target and E_{int} the energy in the plate interior (outside the sponge). The identity is exact. Channeling acts on G_φ and does not depend on frequency, because (6) is non-dispersive. The field-growth factor depends on standing-wave interference in the finite, lightly damped plate, and so on frequency. A gain beyond G_φ at the design frequency should therefore not survive a frequency change.
- **W3 (flux follows fibers).** In any slowly curving fiber field, optimized or not, the local energy flux should align with the local fiber director far better than with the drive direction. The control is a set of curved fields that were never optimized: if channeling were selected by the optimizer rather than imposed by the lamina, the flux would not follow their fibers.
- **W4 (anisotropy).** G_φ should grow with the anisotropy, following s_{20}, from 1 at isotropy.
- **W5 (limits).** A tow of minimum radius R turns by at most ℓ/R over a run ℓ. Energy channeled along the fibers can therefore be redirected only as fast as the fibers turn. Redirection should survive while tows can still turn on the wavelength scale, and fail once R exceeds it. Exact cloaking, by contrast, needs the singular anisotropy of transformation media [14]. A fixed-contrast lamina must therefore rely on interference to cancel a scattered field, and we expect such cancellation to be tuned to one frequency (a hypothesis, tested in Sec. 6.5, not a theorem).

4 Finite-element discretization in `dolfinx`

4.1 Function spaces, element types, and the mesh for each problem

Structured and boundary-conforming meshes both appear, so we state the discrete spaces explicitly before deriving anything. Table 1 collects them.

The state space for the wave-control problems is a *four-component* continuous Lagrange space,

$$\mathsf{U}_h \in \mathcal{W}_h = \big[\mathbb{P}_1(\mathcal{T}_h)\big]^4 = \mathrm{span}\{\texttt{("Lagrange",1,(4,))}\}, \qquad \mathsf{U} = (u_x^r, u_y^r, u_x^i, u_y^i), \tag{8}$$

i.e. *four* scalar degrees of freedom per vertex: the two in-plane components of the real part and the two of the imaginary part. This is the discrete carrier of the real four-field split (11) — it is why a real `PETSc` build suffices, and it is the reason the block operator is 4×4 rather than 2×2.

The design fields live in the *discontinuous, element-wise constant* space

$$\theta_h \in \mathcal{S}_h = \mathbb{DG}_0(\mathcal{T}_h) = \mathrm{span}\{\texttt{("DG",0)}\}, \tag{9}$$

one orientation per cell. The pairing $\mathcal{W}_h = [\mathbb{P}_1]^4$ with $\mathcal{S}_h = \mathbb{DG}_0$ is deliberate: the state must be conforming in H^1, whereas the design must be free to vary cell-by-cell without any inter-element continuity, and the cell gradient (19) is then obtained simply by differentiating the residual against the $\mathbb{DG}_0$ test function.

The adjoint is solved in the same space. We use a Galerkin discretization, so the adjoint state $\mathbf{\Lambda}_h \in \mathcal{W}_h$ — the *same* $[\mathbb{P}_1]^4$ space as the forward state, on the same mesh. Discretely the adjoint operator is the transpose $\mathcal{A}_h^{\mathsf{T}}$ of the forward matrix, not a separately discretized continuous adjoint; this "discretize-then-differentiate" order is what makes the gradient exact for the discrete objective (and is why it matches finite differences to $\sim 10^{-6}$ rather than to truncation error). Practically, the forward solve factors $\mathcal{A}_h = \mathbf{LU}$ and the adjoint solves with the transposed operator $\mathcal{A}_h^{\mathsf{T}}$ by a second sparse LU factorization, so one gradient costs two factorizations (the forward factors could be reused for the transposed solve; we do not exploit this).

Meshes. The lens uses a structured right-triangle mesh of $[0, L_x] \times [0, L_y]$. The cloak and the panel instead use a `gmsh`/OCC mesh in which each circular hole is *cut from the geometry* so the triangulation *conforms* to the circle, refined on the rim. With linear (P1) elements the resolution of the slowest (quasi-shear) wave is about 9 elements per wavelength for the lens ($\omega = 40$), 4.9 for the cloak ($h = 0.045$, $\omega = 50$) and 4.0 for the panel ($h = 0.055$, $\omega = 50$); the coarser two are checked by refinement in Appendix F. The hole boundary is then exactly circular and traction-free by the natural boundary condition of the weak form. We flag the distinction because it is the substantive reason for using `dolfinx` here at all: representing a hole instead as a soft $\mathbb{DG}_0$ density void on a structured grid staircases the boundary to within one element, scatters off the staircase corners, and blurs the very radius R that the curvature rule $\zeta_{\mathrm{all}} = 1/R$ is tied to.

4.2 Real four-field split of the complex vector problem

The weak form of (4) is the complex-symmetric system

$$\big[(1+\mathrm{i}\eta)\mathbf{K}(\boldsymbol{\theta}) - \omega^2\mathbf{M} + \mathrm{i}\omega\mathbf{C}\big]\mathbf{u} = \mathbf{f}, \quad \mathbf{K} = \int_\Omega \boldsymbol{\varepsilon}(\cdot)^{\mathsf{T}}\mathbf{D}_f(\theta)\boldsymbol{\varepsilon}(\cdot), \;\; \mathbf{M} = \int_\Omega \rho\,(\cdot)\cdot(\cdot), \;\; \mathbf{C} = \int_\Omega c\,(\cdot)\cdot(\cdot). \tag{10}$$

Table 1: Discrete spaces, element types and meshes ($\mathcal{T}_h$ the cell complex).

	wave control (`dolfinx`)
cells	triangles ($\mathbb{P}_1$ affine)
mesh	structured; `gmsh` conforming w/ holes
state space	$[\mathbb{P}_1]^4$, `("Lagrange",1,(4,))`
DOF per node	4 ($u_x^r, u_y^r, u_x^i, u_y^i$)
DOF per element	12
quadrature	UFL-estimated degree
mass	consistent
design θ	$\mathbb{DG}_0$, `("DG",0)`
parametrization	CS-RBF $\theta = \mathbf{B}\hat{\theta}$
solve	sparse LU (real `PETSc`)
dual	adjoint in $\mathcal{W}_h$, $\mathcal{A}_h^{\mathsf{T}}$ (second sparse LU)

The `PETSc` build stores *real* scalars only, so we split $\mathbf{u} = \mathbf{u}_r + \mathrm{i}\mathbf{u}_i$ (each a 2-vector) and solve the equivalent real block system on the four-component field $U = (u_{xr}, u_{yr}, u_{xi}, u_{yi})$,

$$\begin{bmatrix} A & -B \\ B & A \end{bmatrix} \begin{bmatrix} \mathbf{u}_r \\ \mathbf{u}_i \end{bmatrix} = \begin{bmatrix} \mathbf{f}_r \\ \mathbf{f}_i \end{bmatrix}, \qquad A = \mathbf{K}(\boldsymbol{\theta}) - \omega^2 \mathbf{M}, \qquad B = \eta\, \mathbf{K}(\boldsymbol{\theta}) + \omega \mathbf{C}, \tag{11}$$

which is exactly $[\mathrm{Re}, -\mathrm{Im}; \mathrm{Im}, \mathrm{Re}]$ of the complex operator (the structural damping $\eta\mathbf{K}$ joins the sponge in the imaginary block B).

Where (11) comes from. Write the complex operator as $\mathcal{A}_{\mathbb{C}} = A + \mathrm{i}B$ with $A = \mathbf{K} - \omega^2\mathbf{M}$ and $B = \eta\mathbf{K} + \omega\mathbf{C}$ *both real and symmetric*. Expanding the complex product and separating real and imaginary parts,

$$(A + \mathrm{i}B)(\mathbf{u}_r + \mathrm{i}\mathbf{u}_i) = \underbrace{\left(A\mathbf{u}_r - B\mathbf{u}_i\right)}_{\text{real part}} + \mathrm{i}\,\underbrace{\left(B\mathbf{u}_r + A\mathbf{u}_i\right)}_{\text{imaginary part}} \overset{!}{=} \mathbf{f}_r + \mathrm{i}\mathbf{f}_i, \tag{12}$$

and equating the two parts *separately* — legitimate because $\{1, \mathrm{i}\}$ is a basis of $\mathbb{C}$ over $\mathbb{R}$ — gives the two coupled real equations $A\mathbf{u}_r - B\mathbf{u}_i = \mathbf{f}_r$ and $B\mathbf{u}_r + A\mathbf{u}_i = \mathbf{f}_i$, i.e. exactly (11). This is the standard *realification*, the injective ring homomorphism $a + \mathrm{i}b \mapsto \left[\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right]$ applied blockwise; nothing is approximated, dropped, or symmetrized.

Proposition 2 (Exactness of the split and of its adjoint)**.** *Let $A, B \in \mathbb{R}^{n\times n}$ be symmetric and $\mathcal{A} = \left[\begin{smallmatrix} A & -B \\ B & A \end{smallmatrix}\right]$. Then*

1. *$(\mathbf{u}_r, \mathbf{u}_i)$ solves $\mathcal{A}(\mathbf{u}_r, \mathbf{u}_i)^{\mathsf{T}} = (\mathbf{f}_r, \mathbf{f}_i)^{\mathsf{T}}$ if and only if $\mathbf{u} = \mathbf{u}_r + \mathrm{i}\mathbf{u}_i$ solves $(A + \mathrm{i}B)\mathbf{u} = \mathbf{f}_r + \mathrm{i}\mathbf{f}_i$; in particular $\mathcal{A}$ is invertible exactly when $A + \mathrm{i}B$ is.*
2. *$\mathcal{A}^{\mathsf{T}} = \left[\begin{smallmatrix} A & B \\ -B & A \end{smallmatrix}\right]$ is the realification of $A - \mathrm{i}B = (A + \mathrm{i}B)^{\mathsf{H}}$.*

Proof. (1) is the displayed computation read in both directions. For (2), transposing blockwise and using $A^{\mathsf{T}} = A$, $B^{\mathsf{T}} = B$ gives $\mathcal{A}^{\mathsf{T}} = \left[\begin{smallmatrix} A & B \\ -B & A \end{smallmatrix}\right]$, which is the image of $A - \mathrm{i}B$ under $a + \mathrm{i}b \mapsto \left[\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right]$; and $(A + \mathrm{i}B)^{\mathsf{H}} = A^{\mathsf{H}} - \mathrm{i}B^{\mathsf{H}} = A - \mathrm{i}B$ since A, B are real symmetric. □

Part (2) is the reason the real formulation costs us nothing in the optimization: the discrete adjoint equation (18) is solved with the *transpose* $\mathcal{A}^{\mathsf{T}}$, and Proposition 2 says that transposing the

real block system is precisely applying the complex *Hermitian* adjoint. So the real solve returns the true complex adjoint state, not an approximation of it — which is why the resulting gradient matches finite differences to $\sim 10^{-6}$.

Remark 2 (Cost, and when the split can be dropped)**.** *$\mathcal{A}$ is real but* non-*symmetric whenever $B \neq 0$ (i.e. whenever there is damping or a sponge), so it is factored by sparse LU, not Cholesky. The split doubles the unknowns; for a sparse factorization this costs a small constant factor over a native complex build of half the size, which we accept because the split is exact and runs on a real-scalar* `PETSc` *build. With a complex-scalar* `PETSc/SLEPc` *the entire construction of this subsection collapses to assembling* (10) *directly on a two-component complex space, and only the objective (which is already real-valued) needs care. Nothing else in the paper would change: the adjoint, the sensitivity* (19) *and every reported number are properties of the discrete problem, not of how its scalars are stored.*

We discretize U on the four-component continuous P_1 space (8) and assemble (11) as a single bilinear form. Writing the stiffness pairing $s(\mathbf{a},\mathbf{b}) = \int_\Omega \boldsymbol{\varepsilon}(\mathbf{a})^{\mathsf{T}}\mathbf{D}_f(\theta)\boldsymbol{\varepsilon}(\mathbf{b})$ and the mass/sponge pairings $m(\mathbf{a},\mathbf{b}) = \int_\Omega \rho\,\mathbf{a}\cdot\mathbf{b}$, $d(\mathbf{a},\mathbf{b}) = \int_\Omega c\,\mathbf{a}\cdot\mathbf{b}$ for the test 2-vectors $\mathbf{p}_r, \mathbf{p}_i$,

$$\begin{aligned} a(U,P;\boldsymbol{\theta}) &= s(\mathbf{u}_r,\mathbf{p}_r) - \omega^2 m(\mathbf{u}_r,\mathbf{p}_r) - \eta\, s(\mathbf{u}_i,\mathbf{p}_r) - \omega\, d(\mathbf{u}_i,\mathbf{p}_r) \\ &\quad + s(\mathbf{u}_i,\mathbf{p}_i) - \omega^2 m(\mathbf{u}_i,\mathbf{p}_i) + \eta\, s(\mathbf{u}_r,\mathbf{p}_i) + \omega\, d(\mathbf{u}_r,\mathbf{p}_i), \qquad \ell(P) = \int_\Omega \mathbf{f}\cdot\mathbf{p}_r. \end{aligned} \tag{13}$$

The orientation θ, the sponge c and the region indicators are element-wise (DG_0) fields; $\mathbf{D}_f(\theta)$ is built from (2) directly in UFL. Material, thickness and density are the same everywhere and in every design: voids are cut from the mesh, never represented by a density. We assemble (13) and solve with a sparse LU factorization; write the discrete system $\mathcal{A}(\boldsymbol{\theta})\,\mathsf{U} = \mathsf{b}$.

4.3 Absorbing sponge and drive

The sponge coefficient ramps quadratically into a boundary margin of width m,

$$c(\mathbf{x}) = c_0\left[\max\left(0,\ \tfrac{x-(L_x-m)}{m},\ \tfrac{m-y}{m},\ \tfrac{y-(L_y-m)}{m}\right)\right]^2, \tag{14}$$

absorbing outgoing waves at the right, top and bottom edges (the left edge carries the drive). The layer is imperfect: for the along-fiber qP wave it reflects $|R| = 0.31$–0.34 at the lens frequencies, and $|R| \leq 0.13$ when its width is doubled (Appendix F), and the finite plate is resonant (the resonant growth persists with the better absorber and scales as $1/\eta$, Appendix F, so it is not set by the absorber alone). The drive is an x-directed line source concentrated near the left edge,

$$\mathbf{f}(\mathbf{x}) = \left(A\,e^{-\beta(x-x_0)^2},\ 0\right), \qquad (A,\beta,x_0) = (30,\ 80,\ 0.1), \tag{15}$$

a plane pressure wave. Parameters (c_0, m) are $(80, 0.6)$ for the lens and $(70, 0.6)$ for the cloak (Table 7). Since $\mathbf{C}, \mathbf{M}$ and $\mathbf{f}$ are θ-independent, they drop out of every design sensitivity (§5).

5 Objectives and the discrete-adjoint sensitivity

Both objectives are quadratic in the state, $J = \int_\Omega w\,\Psi(U)$ with w a region indicator:

$$\Psi_{\text{lens}} = |\mathbf{u}|^2 = \sum_{k=0}^{3} U_k^2, \qquad \Psi_{\text{cloak}} = |\mathbf{u}-\mathbf{u}_{\text{ref}}|^2 = \sum_{k=0}^{3} (U_k - U_k^{\text{ref}})^2, \tag{16}$$

the focus energy in a target box (maximized) and the far-field mismatch to a void-free reference (minimized).

5.1 Lagrangian, adjoint equation, and residual derivative

Adjoin the forward problem $\mathcal{A}(\boldsymbol{\theta})\mathsf{U} - \mathsf{b} = \mathbf{0}$ with a multiplier $\boldsymbol{\Lambda}$: $\mathcal{L} = J(\mathsf{U}) + \boldsymbol{\Lambda}^{\mathsf{T}}(\mathcal{A}\mathsf{U} - \mathsf{b})$. On the constraint manifold $\mathcal{L} = J$, and expanding the total derivative,

$$\frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \underbrace{\left(\tfrac{\partial J}{\partial \mathsf{U}} + \mathcal{A}^{\mathsf{T}}\boldsymbol{\Lambda}\right)^{\mathsf{T}} \tfrac{\mathrm{d}\mathsf{U}}{\mathrm{d}\theta_\ell}}_{\text{eliminated by the adjoint}} + \boldsymbol{\Lambda}^{\mathsf{T}} \frac{\partial \mathcal{A}}{\partial \theta_\ell}\mathsf{U}. \tag{17}$$

Choosing $\boldsymbol{\Lambda}$ to kill the implicit state term gives the *discrete adjoint equation*, and the design derivative then reduces to the explicit θ-dependence of the residual:

$$\boxed{\mathcal{A}(\boldsymbol{\theta})^{\mathsf{T}}\boldsymbol{\Lambda} = -\frac{\partial J}{\partial \mathsf{U}}} \qquad \boxed{\frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \frac{\partial R}{\partial \theta_\ell}, \quad R(\boldsymbol{\theta}) \equiv a(\mathsf{U}, \boldsymbol{\Lambda}; \boldsymbol{\theta}) - \ell(\boldsymbol{\Lambda})} \tag{18}$$

with $\mathsf{U}, \boldsymbol{\Lambda}$ frozen. Only the stiffness terms in R carry θ, so

$$\frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \int_{\Omega_\ell} \Big[\boldsymbol{\varepsilon}(\mathbf{u}_r)^{\mathsf{T}} \tfrac{\partial \mathbf{D}_f}{\partial \theta} \boldsymbol{\varepsilon}(\boldsymbol{\lambda}_r) + \boldsymbol{\varepsilon}(\mathbf{u}_i)^{\mathsf{T}} \tfrac{\partial \mathbf{D}_f}{\partial \theta} \boldsymbol{\varepsilon}(\boldsymbol{\lambda}_i) + \eta \big(\boldsymbol{\varepsilon}(\mathbf{u}_r)^{\mathsf{T}} \tfrac{\partial \mathbf{D}_f}{\partial \theta} \boldsymbol{\varepsilon}(\boldsymbol{\lambda}_i) - \boldsymbol{\varepsilon}(\mathbf{u}_i)^{\mathsf{T}} \tfrac{\partial \mathbf{D}_f}{\partial \theta} \boldsymbol{\varepsilon}(\boldsymbol{\lambda}_r) \big) \Big], \tag{19}$$

where $\boldsymbol{\lambda}_r, \boldsymbol{\lambda}_i$ are the real/imag adjoint displacements and $\partial \mathbf{D}_f / \partial \theta = \mathbf{T}' \mathbf{Q}_f \mathbf{T}^{\mathsf{T}} + \mathbf{T} \mathbf{Q}_f \mathbf{T}'^{\mathsf{T}}$. The mass, sponge and source carry no θ-dependence and drop out: the whole gradient is built from $\partial \mathbf{D}_f / \partial \theta$, so it vanishes identically for an isotropic material, making Remark 1 an exact statement about the gradient.

5.2 Automatic implementation, scaling, and verification

Each boxed step is one line of UFL. The adjoint load $\partial J / \partial \mathsf{U}$ is `derivative`$(w\Psi(U)\,\mathrm{d}\mathbf{x}, U, P)$; (18) is solved with the transposed operator $\mathcal{A}^{\mathsf{T}}$ (a second sparse LU factorization); the cell gradient is `derivative`$(R, \theta, \hat{P})$ against the DG_0 test function — no element derivatives by hand. Both the lens and cloak gradients match central finite differences to a relative error $\sim 10^{-6}$. The raw objectives are tiny ($\sim 10^{-4}$), so for the lens we descend the scale-invariant *log*-objective $g_\ell = (\partial_{\theta_\ell} J)/J$, restoring $O(1)$ gradients for the Method of Moving Asymptotes (MMA) [35] under $\theta_\ell \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

5.3 Smooth orientation parametrization and manufacturable toolpaths

A raw per-cell orientation is mesh-scale noisy — un-manufacturable and, driving the wave field, incoherent. We therefore parametrize θ by a coarse grid of support-point angles $\hat{\theta} = \{\hat{\theta}_i\}$ at sites $\{\mathbf{s}_i\}$ (spacing h_s) interpolated to the cell centroids $\{\mathbf{x}_c\}$ with a compactly supported Wendland C^2 radial basis [36] $\phi(q) = (1-q)^4(4q+1)$ (support $q < 1$), as in [11]:

$$\theta_c = (B\hat{\theta})_c = \sum_i B_{ci}\hat{\theta}_i, \qquad B_{ci} = \frac{\phi(q_{ci})}{\sum_j \phi(q_{cj})}, \qquad q_{ci} = \frac{\|\mathbf{x}_c - \mathbf{s}_i\|}{r}, \tag{20}$$

a row-stochastic map ($\sum_i B_{ci} = 1$); B^{T} propagates to the gradient, $\partial J / \partial \hat{\theta} = B^{\mathsf{T}}(\partial J / \partial \theta)$. Support spacing and radius are $(h_s, r) = (0.33, 0.7)$ for the lens and $(0.28, 0.6)$ for the cloak. This CS-RBF map makes the design low-dimensional and smooth *by construction*. When the problem has a mirror symmetry (as the lens does about $y = L_y/2$) we enforce it on the design: a fiber field is symmetric under $y \mapsto L_y - y$ iff $\theta(x, L_y - y) = -\theta(x, y)$, so we use a support grid symmetric about the axis and project the design each iteration onto that subspace, $\hat{\theta} \mapsto \frac{1}{2}(\hat{\theta} - \hat{\theta}_{\text{mirror}})$. This

removes the spurious asymmetry that the (non-symmetric) triangulation and optimizer would otherwise introduce, yielding a symmetric lens with a horizontal fiber on the axis. We delineate the resulting fiber *toolpaths* with the wave projection of [11]: from the sign-free double-angle director $(\cos 2\theta, \sin 2\theta)$ we recover a smooth (seam-free) orientation branch by region-growing sign propagation, then solve for a phase field Φ whose gradient is unit and orthogonal to the fibers,

$$\nabla\Phi(\mathbf{x}) = \mathbf{g} \equiv [-\sin\theta,\ \cos\theta]^{\mathsf{T}}, \qquad \Phi = \arg\min_{\Phi} \int_{\Omega} \|\nabla\Phi - \mathbf{g}\|^2, \tag{21}$$

discretized as a sparse least-squares gradient match on the grid (`lsqr`; matching $\nabla\Phi$ *directly* keeps $|\nabla\Phi|$ close to 1 wherever the director field is nearly divergence-free — the normal-equations Poisson form drops the boundary flux and collapses the pitch). The level sets of Φ at increments d — equivalently the peaks of the wave projection $\chi = \frac{1}{2} + \frac{1}{2}\cos(2\pi\Phi/d)$ — are the as-manufactured tow center-lines; they follow the fiber direction, are masked out of any prescribed void, so no toolpath crosses a hole, and their spacing is d only where the field does not converge or diverge: where it converges (as near the axis of the lens, Fig. 2c) the tows crowd together, which a fiber placement process would resolve by cutting or adding tows. These paths visualize the field; we do not model the resulting thickness or gap variations. Together with the structural damping η, this smoothing is what renders the optimized fields coherent rather than speckled.

5.4 Fiber-curvature (curl) constraint

A printable toolpath cannot bend arbitrarily tightly: the tow has a minimum turning radius. Following [11], we bound the local *fiber-path curvature* — the directional derivative of the orientation along the fiber, $\mathbf{n} = (\cos\theta, \sin\theta)$,

$$\zeta(\mathbf{x}) = (\mathbf{n}\cdot\nabla)\theta = \cos\theta\,\partial_x\theta + \sin\theta\,\partial_y\theta, \qquad |\zeta(\mathbf{x})| \le \zeta_{\text{all}}, \tag{22}$$

so ζ is exactly the signed curvature of the fiber streamlines and ζ_{all} is the inverse minimum radius. The spatial derivatives come *analytically* from the CS-RBF map (20): $\partial_x\theta = \mathbf{B}_x\hat{\theta}$, $\partial_y\theta = \mathbf{B}_y\hat{\theta}$. Writing $W_{ci} = \phi(q_{ci})$, $S_c = \sum_j W_{cj}$ (so $B_{ci} = W_{ci}/S_c$) and differentiating the kernel *and* its normalization consistently,

$$\partial_x W_{ci} = -\frac{20(1-q_{ci})^3}{r^2}\,(x_c - s_{ix}), \qquad (\mathbf{B}_x)_{ci} = \frac{\partial_x W_{ci}}{S_c} - B_{ci}\,\frac{\sum_j \partial_x W_{cj}}{S_c}, \tag{23}$$

and likewise $\mathbf{B}_y$ (using $\phi'(q) = -20\,q(1-q)^3$); we impose (22) through a penalty on the cell average of the excess of ζ^2 over ζ_{all}^2,

$$\mathcal{C}(\hat{\theta}) = \left\langle\ \max\!\left(0,\ \zeta^2 - \zeta_{\text{all}}^2\right)\right\rangle, \qquad \min_{\hat{\theta}}\ -\ln E(\hat{\theta}) + \mu\,\mathcal{C}(\hat{\theta}), \tag{24}$$

whose gradient uses the curvature Jacobian $\partial\zeta/\partial\hat{\theta} = \mathrm{diag}(-\sin\theta\,\theta_x + \cos\theta\,\theta_y)\mathbf{B} + \mathrm{diag}(\cos\theta)\mathbf{B}_x + \mathrm{diag}(\sin\theta)\mathbf{B}_y$ (finite-difference verified to $\sim 10^{-10}$), with the penalty weight μ raised by continuation until the constraint is met.

6 Results

All results use the CFRP lamina of Sec. 2 unless stated, an x-directed drive at the left edge (a plane-wave line source for the lens and the cloak, a Gaussian beam for the panel, Table 8), and the absorbing sponge on the other three sides. Only the fiber orientation is designed. Material, thickness and mass are identical to the straight-fiber baseline.

6.1 The lens: what the optimizer buys, and what it does not (W1, W2)

On a 4×3 plate at $\omega = 40$ (Table 7), maximizing the energy in a target box (dashed square in Fig. 2) yields a mirror-symmetric field of tows bending toward the target near the right end of the plate. The single-frequency target-energy gain against straight fibers is $G = 78\times$. The decomposition (7) shows what this number contains (Fig. 3, Table 2).

- **Redirection.** The target's share of the interior energy rises from 1.47% to 8.6%, a redirection $G_\varphi = 5.8\times$.
- **Field growth.** The whole interior field grows $13.4\times$.
- **Source power.** The design draws $4.1\times$ more power from the force source.

Sweeping the drive frequency on the *fixed* design separates the two factors as W2 predicts. The redirection is broadband once the wavelength is short against the lens: $G_\varphi \geq 3.7\times$ for every $\omega \in [37.5, 60]$, and the band average $\langle G_\varphi \rangle_{\text{band}}$, defined throughout as the mean of G_φ at $\omega \in \{35, 40, 45, 50\}$, is $5.2\times$. At $\omega = 20$ and 25 the redirection collapses and even reverses ($G_\varphi < 1$, Table 2): there the along-fiber wavelength exceeds the length over which the fiber field turns, and the ray picture of Sec. 3 no longer applies. These values depend on the absorbing layer: with one reflecting $|R| \leq 0.13$ instead of 0.31–0.34, the lens re-optimized from scratch gives $G = 70\times$ at $\omega = 40$, split into $G_\varphi = 4.4\times$ and $15.9\times$ growth, and $G_\varphi \geq 3.6\times$ over $[40, 60]$ ($2.8\times$ at 37.5), band average $3.9\times$ (Appendix F). The split, and the resonant nature of the excess gain, are unchanged. The field growth is a resonance of the finite plate at the design frequency. At $\omega = 37.5$ and 42.5, only $\pm 6\%$ away, G falls to $6.6\times$ and $10.1\times$, while G_φ changes far less (from 5.8 to 3.7–6.1). A large single-frequency focus gain is therefore not by itself evidence of steering. It mostly measures how well the optimizer tuned a standing wave. The physical, transferable result is the redirection.

For W1, setting the lamina isotropic makes $G = G_\varphi = 1$ to machine precision, and every orientation gradient vanishes identically (Remark 1; Sec. 6.3).

Table 2: The lens design (optimized at $\omega = 40$) evaluated across frequency. G: target-energy gain; G_φ: redirection (target share of interior energy, relative to straight fibers); growth: interior energy ratio; P: injected power ratio. $G = G_\varphi \times$growth exactly.

ω	G	G_φ	growth	P/P_0
20.0	0.2	0.25	0.74	0.61
25.0	0.9	0.41	2.26	0.96
30.0	3.7	1.65	2.25	0.92
35.0	2.1	2.34	0.89	0.49
37.5	6.6	3.74	1.77	0.80
40.0	**78.0**	**5.82**	**13.41**	**4.13**
42.5	10.1	6.10	1.65	0.95
45.0	16.3	6.54	2.49	1.28
50.0	5.5	6.01	0.91	0.70
55.0	5.7	4.27	1.34	0.91
60.0	5.8	3.71	1.58	1.08

6.2 Mechanism: the energy flux follows the fibers (W3)

We compute the time-averaged elastic energy flux of the solved field cell by cell, $P_j = -\frac{1}{2}\,\mathrm{Re}\{\sigma_{ij}\bar{v}_i\} = \frac{\omega}{2}\left[\sigma_{ij}(\mathbf{u}^{\mathrm{r}})\,u_i^{\mathrm{i}} - \sigma_{ij}(\mathbf{u}^{\mathrm{i}})\,u_i^{\mathrm{r}}\right]$ (for $e^{+\mathrm{i}\omega t}$, $\boldsymbol{v} = \mathrm{i}\omega\mathbf{u}$), and measure its angle to the local

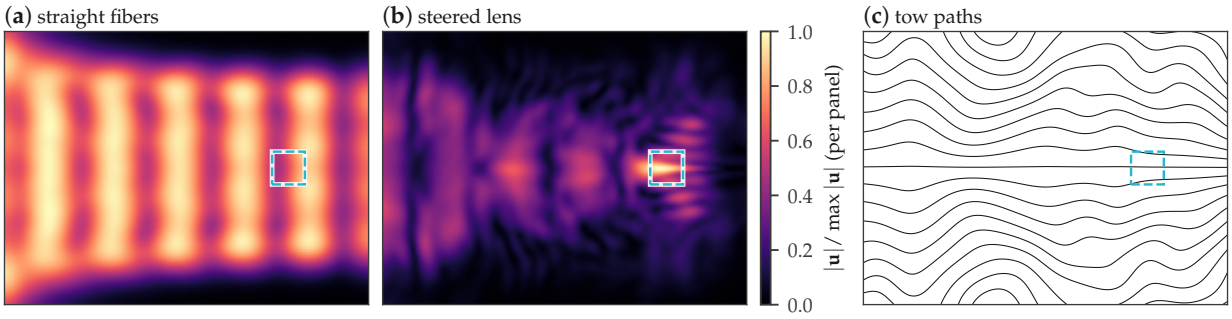


Figure 2: The orientation-only lens at its design frequency ($\omega = 40$), on the full 4×3 plate driven at its left edge. **(a)** Field magnitude $|\mathbf{u}|$ with straight fibers, and **(b)** with the optimized fiber field; each map is normalized by its own maximum, so the comparison is of *where* the energy goes (the absolute ratio is the gain G); dashed square: the objective region $(0.78L_x, L_y/2) \pm 0.18$. **(c)** Optimized tow paths (mirror-symmetric, bending toward the objective region near the right end). The 78× focus gain of (b) over (a) is 5.8× redirection times a resonant field growth (Sec. 6.1).

fiber director (Fig. 4).

- **Optimized lens.** The $|\boldsymbol{P}|$-weighted mean deviation of the flux from the fibers is 6.7°, and 93% of the energy flux runs within 20° of them. The flux deviates 19.2° on average from the drive direction x.
- **Across the band.** At $\omega = 35, 45, 50$ the flux–fiber deviation is 11.6°, 5.5° and 4.5°.
- **Control: fields that were never optimized.** Three random smooth CS-RBF fields (same support grid as the lens, support values drawn from $\mathcal{N}(0, 0.5\,\text{rad})$) and one analytic field, $\theta = 0.6\sin(\pi x/2)\sin(2\pi y/3)$, give flux–fiber deviations of only 1.0–4.2° at $\omega = 40$ and 50, against 9–15° from the drive direction (Appendix F). Their fields turn more gently than the lens's, which is why they channel even more tightly.

The energy goes where the fibers point, at every frequency tested and whether or not the field was optimized: channeling is imposed by the lamina, and the optimizer only chooses where the fibers point. This is the channeling predicted by (6), and it is why the redirection is broadband.

6.3 How the control scales with anisotropy (W4)

Along the family $\mathbf{Q}(s) = \mathbf{Q}_{\text{iso}} + s(\mathbf{Q}_f - \mathbf{Q}_{\text{iso}})$, which holds the isotropic invariants and scales the anisotropic ones by s, we re-optimize the lens for each material (Table 3, Fig. 5a). The band-averaged redirection does not decrease with anisotropy in these runs. It is exactly 1 at isotropy, rises steeply to 4.2× by $E_{f1}/E_{f2} \approx 3$ (where s_{20} is already about 60%), reaches 5.2× for CFRP, and 6.5× at $E_{f1}/E_{f2} = 32$. The single-frequency gain G along the same family is *non-monotone* (24× at $E_{f1}/E_{f2} = 2.9$ but 16× at 5.4), because it is dominated by whichever resonance each optimization happens to tune. The trend should be read coarsely: for the same CFRP problem, different optimizer runs give a band-averaged redirection from 5.2× (the lens of Sec. 6.1) to 6.8× (the best front design, Sec. 6.4), a spread comparable to the steps between neighboring laminae. Robust are the end points (1 at isotropy, several times once $E_{f1}/E_{f2} \gtrsim 3$), not the detailed shape, such as the plateau between $E_{f1}/E_{f2} = 2.9$ and 5.4.

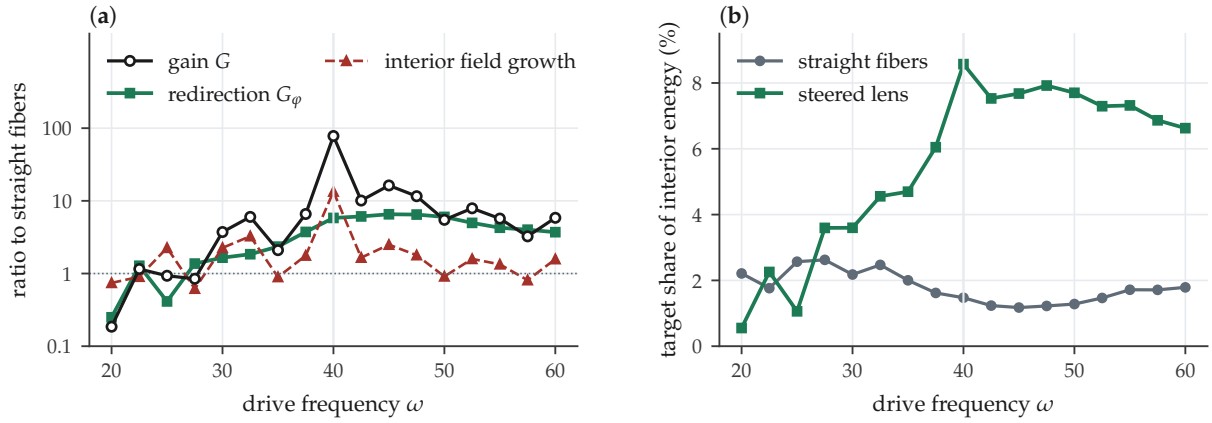


Figure 3: What the lens does across frequency (design fixed, optimized at $\omega = 40$). **(a)** Single-frequency gain G (log scale) and its two factors: redirection G_φ (broadband) and interior field growth (resonant); dotted: parity with straight fibers. **(b)** Target share of interior energy, straight vs. steered: the steered plate sends at least 3.7× more of its energy to the target at every frequency in $\omega \in [37.5, 60]$.

Table 3: Lens re-optimized for each lamina of the anisotropy family. s_{20}: channeling share (Sec. 3).

s	E_{f1}/E_{f2}	s_{20}	$G(40)$	$G_\varphi(40)$	$\langle G_\varphi \rangle_{\text{band}}$
0.00	1.0	22%	1.0	1.00	1.00
0.25	1.7	42%	6.9	2.63	1.87
0.50	2.9	60%	24.1	6.92	4.21
0.75	5.4	75%	16.2	5.49	4.22
1.00	14.6	88%	78.0	5.82	5.18
1.10	31.9	94%	162.4	5.85	6.55

6.4 Manufacturing limits (W5): tow curvature

We bound the tow curvature $|\zeta| \leq \zeta_{\text{all}} = 1/R$, keeping the lens's own mesh, frequency and parametrization (Table 4, Fig. 5b).

- **Plateau.** The best band-averaged redirection stays at 6.4–6.8× for every limit down to $\zeta_{\text{all}} = 1.0$, i.e. a minimum tow radius $R = 1.0 \approx 0.7\lambda$, where $\lambda = 1.42$ is the along-fiber qP wavelength at $\omega = 40$ (the longest wavelength present; against the cross-fiber qP wavelength 0.37 the same radius is 2.7 wavelengths). We use the along-fiber value because the channeled energy travels along the fibers. The front lies at the edge of the slowly-turning regime assumed by the ray argument of Sec. 3, so we read it as an empirical scale rather than a ray prediction.
- **Decline.** Beyond that radius the best design found falls gradually, not abruptly: 4.9× at $R = 1.25$, 3.2× at $R = 1.67$ ($\approx 1.2\lambda$), 2.3× at $R = 2.2$ and 1.75× at $R = 3.33$. Tows that cannot turn on the wavelength scale cannot converge the channeled energy on the target within the plate.

Physically, energy can be redirected only if the tows can turn on the wavelength scale.

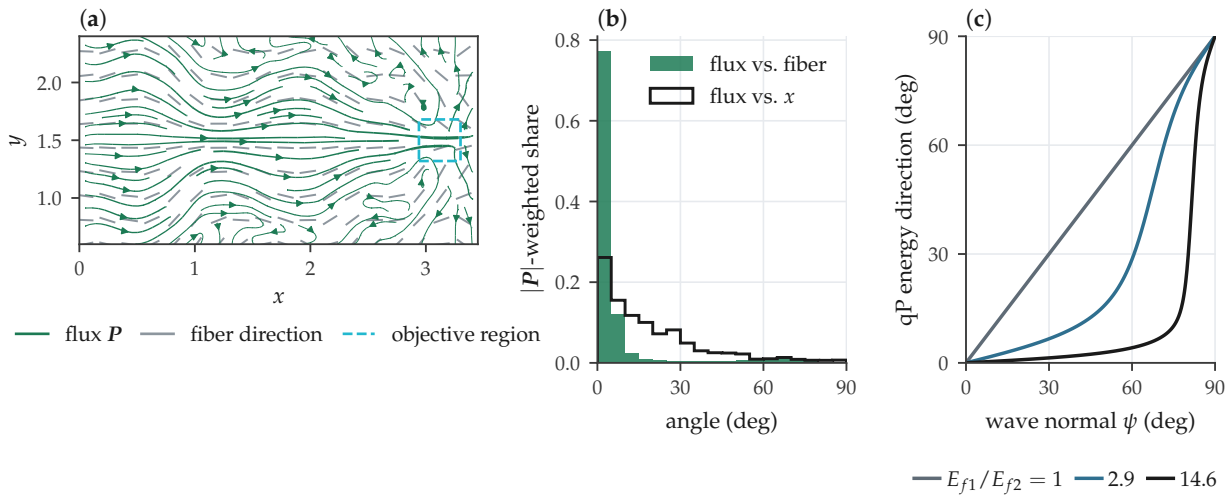


Figure 4: Energy channeling. **(a)** Time-averaged energy-flux streamlines of the optimized lens at $\omega = 40$ over the fiber director field (gray dashes), in the window $0 \le x \le 3.4$, $0.6 \le y \le 2.4$ that excludes the absorbing layers; dashed square: objective region. **(b)** The $|\boldsymbol{P}|$-weighted distribution of the angle between the flux and the local fiber, and between the flux and the drive direction. **(c)** The qP energy direction vs. wave normal (6) for an isotropic lamina (diagonal: no skew), the intermediate lamina $E_{f1}/E_{f2} = 2.9$ of the family $\mathbf{Q}(s)$, and CFRP ($E_{f1}/E_{f2} = 14.6$): in CFRP most normals carry their energy close to the fiber.

How the front is computed, and how robust it is. The optimizations are local and strongly start-dependent under a curvature limit. At each limit from $\zeta_{\text{all}} = 0.3$ to 1.5 we therefore run 8 starts (7 of which lead to distinct designs): every earlier front design and three random fields, each scaled until feasible (`lens_front_robust.py`). The spread is large (at $\zeta_{\text{all}} = 1.0$ from 1.0× to 6.4×; shaded bars in Fig. 5b), so single runs can stall far below what the limit permits: an earlier single-run front reported a collapse to 1.3× at $R = 1.67$, which the multi-start study shows was a stalled run. A design whose actual curvature satisfies a limit is feasible for that limit and for every looser one, so Table 4 reports, at each limit, the best band-averaged redirection over *all* computed designs that satisfy it; for $\zeta_{\text{all}} = \infty$ every design counts, and the best (6.8×) comes from a run at $\zeta_{\text{all}} = 1.5$. These values are lower bounds on what each limit allows. The best unconstrained-objective designs also redirect more than the lens of Sec. 6.1 (5.2×): the single-frequency objective does not determine the redirection uniquely, which is another reason to report and optimize the band-averaged quantity.

6.5 The cloak: what bounded anisotropy can and cannot do (W5)

On an 8×5 plate at $\omega = 50$, a circular traction-free void of radius $R_V = 0.55$ is cut conformally from the mesh. Orientation may change only inside a shell $r < R_{\text{des}} = 1.5$ (exactly straight outside, enforced by a smooth taper carried through the chain rule). The far-field mismatch to the void-free reference is minimized outside $R_C = 1.7$, a region the design cannot touch.

- **At the design frequency.** The optimized shell reduces the mismatch 36× (Fig. 6). The transformation-cloak prescription [12, 14], imposed as fiber *directions* on our fixed-contrast lamina, gives 1.9×, because what it actually requires is a diverging *magnitude* of anisotropy, which no lamina has. Its fiber curvature is also singular at the rim: the radial map sends

Table 4: Lens manufacturability front (same mesh, frequency and parametrization as Sec. 6.1). Each row is the best band-averaged redirection over all computed designs whose curvature satisfies the limit (a lower bound on what the limit allows); the last column is the range over the independent starts run at that limit.

ζ_{all}	$R = 1/\zeta_{\text{all}}$	$\max\lvert\zeta\rvert$	$G(40)$	$G_\varphi(40)$	$\langle G_\varphi\rangle_{\text{band}}$	range over starts
∞	0	1.47	13.2	7.49	6.80	–
4	0.25	1.47	13.2	7.49	6.80	–
2.5	0.40	1.47	13.2	7.49	6.80	–
1.5	0.67	1.47	13.2	7.49	6.80	1.74–6.80
1	1.00	0.98	11.1	7.08	6.43	1.04–6.43
0.8	1.25	0.76	7.3	5.28	4.87	0.61–4.87
0.6	1.67	0.57	4.5	3.42	3.21	0.69–3.21
0.45	2.22	0.43	3.1	2.48	2.31	0.76–2.31
0.3	3.33	0.28	2.2	1.89	1.75	0.84–1.75

straight tows onto curves that wrap the void ever more tightly, so no minimum tow radius is respected (its tow paths are not shown; the field in Fig. 6d is what these directions achieve).

- **Mesh sensitivity.** An interference cancellation is sensitive to the discrete dispersion of the mesh it was tuned on. Evaluated on a finer conforming mesh ($h = 0.03$ instead of 0.045), the same design gives 26×, while the curvature-limited and transformation-direction designs change less (9.6 → 8.2×, 1.9 → 2.0×). Re-optimized on the finer mesh (warm start, 80 iterations), the shell reaches 40×, and it is just as narrowband (7.4× and 14.6× at ±5%). The capability therefore persists under refinement, while any one optimized design is tuned to its discretization (Appendix F).
- **Across frequency.** As W5 predicts, the optimized cloak is narrowband (Table 5): 7.5× at $\omega = 47.5$, 14.9× at 52.5, and 1.9× at 40. There it is outperformed by the transformation-direction field (4.6×).
- **Under a tow-curvature limit.** With the curvature bounded at the void's own scale, $\zeta_{\text{all}} = 1/R_V$, the cloak reaches 9.6× at the design frequency (an independent run of the curvature front, Fig. 5c, gives 9.0× at the same limit). This is less peaked but also less narrowband (5.7–6.2× over ±5%, and at least 2.9× over $\omega \in [40, 60]$).

Bounded-anisotropy cloaking is therefore real but tuned. Manufacturable orientation fields can cancel a scatterer's far field strongly at a chosen frequency, and only moderately across a band.

Table 5: Far-field mismatch reduction of three cloak fields across frequency (all on the same conforming mesh; designs fixed).

ω	optimized	curvature-limited ($1/R_V$)	transformation directions
40.0	1.9	3.7	4.6
45.0	2.5	2.9	1.8
47.5	7.5	6.2	2.8
50.0	**36.2**	**9.6**	**1.9**
52.5	14.9	5.7	3.9
55.0	6.4	4.6	2.6
60.0	2.7	2.9	1.1

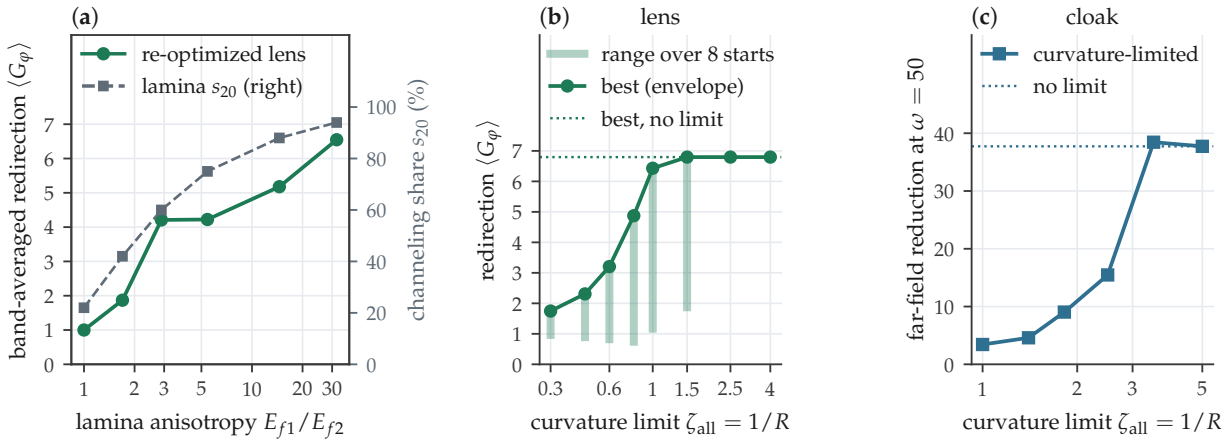


Figure 5: What bounds orientation control. **(a)** Band-averaged redirection of the re-optimized lens vs. lamina anisotropy, with the channeling share s_{20} of the lamina (right axis; both axes start at zero). **(b)** Band-averaged redirection of the lens and **(c)** far-field reduction of the cloak at $\omega = 50$ vs. the tow-curvature limit $\zeta_{\text{all}} = 1/R$ (lengths in the nondimensional units of the plate). In (b), circles: the best design found at each limit over all computed designs that satisfy it (a lower bound on what the limit allows); shaded bars: the range over the independent starts run at that limit; dotted: the best design found with no limit (6.8×, from a run at $\zeta_{\text{all}} = 1.5$). In (c), dotted: the loosest-limit run of this front study ($\zeta_{\text{all}} = 5$, achieved $\max|\zeta| = 4.5$, i.e. effectively unconstrained; 37.7×, while the design of Fig. 6, 36×, is an independent run of the same problem). At $\zeta_{\text{all}} = 3.5$ the constrained cloak (38.4×) slightly exceeds the unconstrained one (37.7×): the optimizer is local, and the warm-started constrained run found a marginally better optimum, so the two are equal within optimizer variability.

6.6 Design practice: a band-robust multifunctional panel

The preceding results imply two design rules. Optimize over a frequency band rather than at one frequency, so that the optimizer cannot buy performance with a resonance. Ask the fiber field to route energy, which channeling does broadband. A bolted 7×4 CFRP panel with four conforming fastener holes is designed this way. A single mirror-symmetric, curvature-feasible field ($\max|\zeta| = 5.4 \leq 1/R_{\text{tow}}$), optimized over 5 frequencies in $[44, 56]$, does four things at the band center (Fig. 7; parameters in Table 8):

- it routes the incident vibration onto a sensor patch (1.5×);
- it shields a payload core to 0.23× its straight-fiber energy;
- it keeps the four fatigue-critical fastener rims at 0.15×;
- it stiffens the bolted mount 1.18× at no added mass (stiffness is an explicit, weighted term of the objective).

Across the band (Fig. 7d) the payload stays at or below 0.32× and the rims at or below 0.49× their straight-fiber energies, and the sensor stays above its straight-fiber level (lowest 1.05× at $\omega = 53$); outside the band it can dip below it (0.93×). A design optimized at the band center alone, with the same settings, routes slightly better (sensor at least 1.18× in the band) and shields better at the band center, but lets the rim protection degrade toward the lower band edge, to 1.09× the straight-fiber energy, i.e. worse than straight fibers. The band objective buys robust rim protection

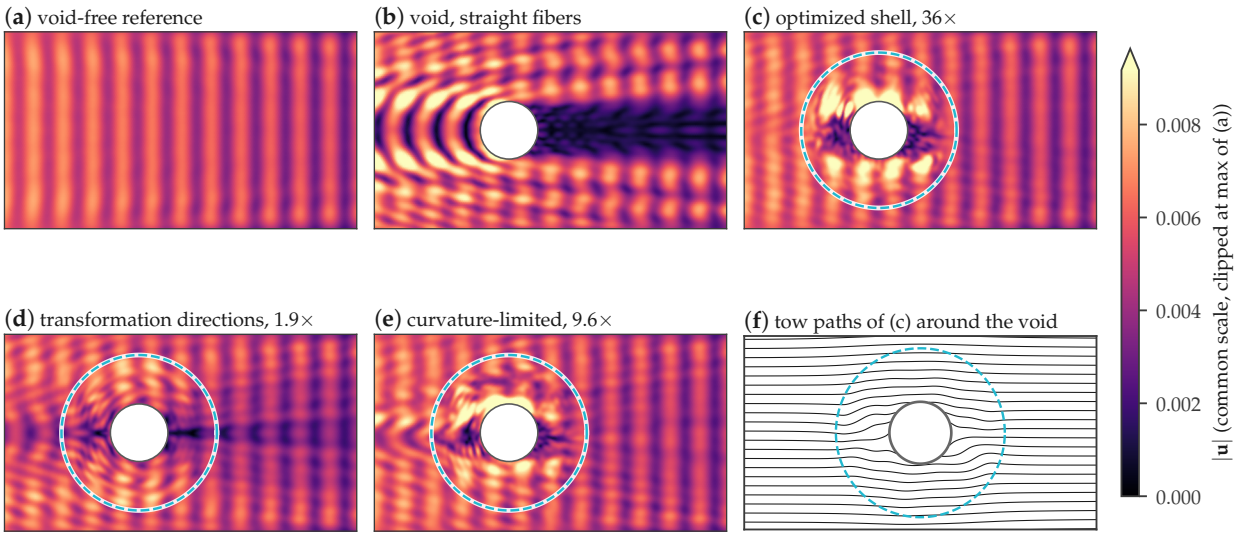


Figure 6: Cloaking a conforming traction-free void ($R_V = 0.55$) at $\omega = 50$, field magnitude $|\mathbf{u}|$ (nondimensional) in the plate interior on one common color scale, clipped at the maximum of the reference (a) (the arrow marks values above it: at most 3.5% of all mesh nodes, near the void, by at most 1.8×). **(a)** Void-free reference. **(b)** The void with straight fibers: shadow and reflections. **(c)** Optimized orientation, free only inside the design shell (dashed circle, same in (c)–(f)). **(d)** Transformation-cloak fiber *directions* imposed on the fixed-contrast lamina. **(e)** Optimized under the tow-curvature limit $\zeta_{\text{all}} = 1/R_V$. **(f)** Tow paths of (c) around the void (same aspect ratio, centered on the void and enlarged by about 7%), wrapping the void inside the shell and straight outside it. Reductions of the far-field mismatch are given in the panel titles; Table 5 shows that the strong reductions are confined to the design frequency, while the curvature-limited design holds a moderate reduction across the band.

at a small cost in routing. The objective, a single signed region weight with logarithmic delivery and straight-fiber normalization, is given in Appendix E.

7 Discussion

7.1 Answer to the question

- **Mechanism.** Fiber orientation controls in-plane elastic waves by *channeling* their energy along the fibers. The control is the redirection of a plate's vibration energy toward a target, by a few times (3.9–5.2× band-averaged for the lens, depending on the absorbing layer), over a broad band.
- **What bounds it.** The lamina anisotropy (none for isotropic material, growing with the channeling share) and the tow curvature (no loss while the minimum tow radius is below about 0.7 along-fiber wavelengths, then a gradual decline of the best design found).
- **Cancellation.** Canceling the field of a finite scatterer (a full cloak) has to rely on interference, because exact cloaking needs anisotropy that no lamina has, and strong cancellation holds only near the design frequency (the curvature-limited design keeps a moderate 2.9–9.6×

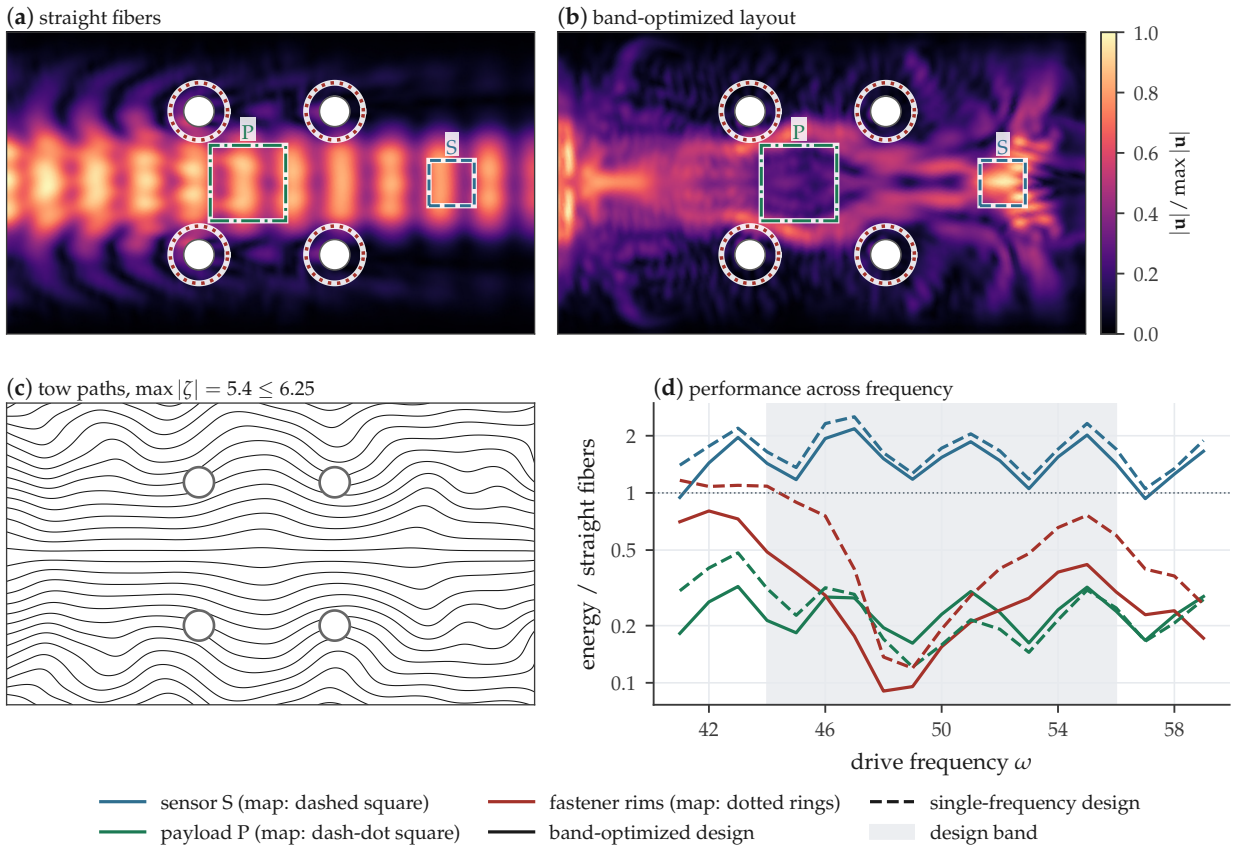


Figure 7: Band-robust multifunctional panel (four conforming fastener holes; drive at the left edge). **(a,b)** $|\mathbf{u}|$ at the band center with straight fibers and with the band-optimized layout, each normalized by its own maximum: the beam splits around the payload (P, green dash-dot) and reconverges on the sensor (S, blue dashed) while the fastener rims (red dotted) stay quiet. **(c)** Tow paths, curvature-feasible. **(d)** Energy in the sensor, payload and rims relative to straight fibers across frequency, for the band-optimized design (solid) and a design optimized at $\omega = 50$ only (dashed); shaded: design band; dotted: parity with straight fibers. The single-frequency design lets rim protection degrade toward the lower band edge.

across $\omega \in [40, 60]$); the inextensible-fiber limit [16] and ground-plane cloaks [17, 18] escape this bound.

7.2 Consequences for how wave-control designs are reported

A finite, lightly damped plate driven at one frequency is a resonator. An optimizer maximizing a field energy at that frequency exploits the resonance whenever it can. The lens shows the size of the effect: 78× at the design frequency, of which 13.4× is field growth, and 6.6× only 6% away. We therefore reaffirm three practices, established for wave topology optimization [29, 28], applied in this paper, and needed before single-frequency gains are compared across studies:

1. report redirection (a share of energy) alongside, or instead of, raw gains;
2. evaluate every design across a band;

Table 6: Status of the predictions.

prediction	evidence	verdict
W1 isotropic plate: no control (identity; consistency check)	Remark 1; Table 3 ($s = 0$)	holds (exact)
W2 redirection is broadband, extra gain is resonant	Table 2, Fig. 3; damping test (App. F)	holds for $\omega \geq 37.5$ ($G_\varphi = 2.3$ at 35); fails at long wavelength ($G_\varphi < 1$ at $\omega = 20$ and 25)
W3 energy flux follows the fibers	lens: 4.5–11.6° across the band (Fig. 4); unoptimized controls: 1.0–4.2°	holds
W4 redirection grows with anisotropy	Table 3, Fig. 5a	holds coarsely (1 at isotropy, several times once $E_{f1}/E_{f2} \gtrsim 3$); finer steps are within optimizer variability
W5 redirection is unaffected while $R \lesssim 0.7\lambda$, then declines; strong cloaking is narrowband	Tables 4, 5; multi-start front	partly holds: redirection is bounded by the tow radius, but declines gradually rather than failing at $R \sim \lambda$ (front values are lower bounds); the cloak half holds (strong cancellation only near the design frequency)

3. optimize band-averaged objectives when the application has a band (Sec. 6.6).

7.3 Limitations and scope

- **Model.** The model is a single plane-stress lamina, with no through-thickness or flexural waves, no tow gaps or overlaps, and constant thickness.
- **Absorbing layer.** The sponge reflects $|R| \approx 0.31$–0.34 of the along-fiber qP wave, so finite-plate standing waves remain. With a sponge twice as wide ($|R| \leq 0.13$) the resonant growth of the lens persists (14.6× at $\omega = 40$), but the redirection is lower: 4.4× at $\omega = 40$ and 3.9× band-averaged, against 5.8× and 5.2× (Appendix F). This is a systematic bias, not scatter: reflections from the layer raise the target share. Re-optimizing the lens with the better absorber does not recover the difference (band-averaged 3.9×, against 3.9× for the original design with that absorber, and $70\times = 4.4\times$ redirection $\times$ 15.9× growth at $\omega = 40$). Absolute redirection values are therefore about a quarter lower with a near-non-reflecting boundary; the conclusions, which rest on the split between broadband redirection and resonant growth, do not change.
- **Discretization.** Lens metrics change by at most about 7% under 2× mesh refinement. The panel's sensor and payload ratios change by at most 6% on a finer mesh, its rim ratios by up to 0.10 in absolute terms. The optimized cloak is mesh-tuned ($36\times \rightarrow 26\times$ on a finer mesh), but the capability persists (40× re-optimized there); we checked two meshes, not a converged sequence.
- **Optimality.** The optimizations are local (MMA, from straight fibers unless stated; the front also uses warm and random starts). The dose and front studies re-optimize each point independently, so their trends carry optimizer noise, and the front values are lower bounds on what each limit allows.
- **Ray picture.** Channeling is argued from plane-wave theory and confirmed by the flux field. A full ray-tracing prediction of the redirection is left for future work.

7.4 Outlook

The channeling picture suggests where orientation design should go next:

- routing problems around prescribed holes and joints, where tows must wrap obstacles within the curvature limit;
- multi-objective layouts that trade wave routing against stiffness;
- combined topology and toolpath design, in which voids are cut rather than softened so that energy cannot pool in compliant regions;
- a ray-theoretic predictor of redirection from the fiber field alone.

8 Conclusions

1. In a CFRP lamina, wave energy is skewed toward the fibers by up to 65°, and 88% of quasi-P wave normals carry their energy within 20° of the fiber direction (Sec. 3).
2. An orientation-only lens redirects energy into its target by 5.8× at its design frequency and by at least 3.7× over $\omega \in [37.5, 60]$; with a better absorbing layer, re-optimized, by 4.4× and at least 3.6× over $[40, 60]$. Its 78× single-frequency gain is mostly a resonant growth of the whole field.
3. In the optimized lens, the energy flux follows the local fibers: mean deviation 6.7° at the design frequency and 4.5–11.6° across the band, and the same holds in curved fields that were never optimized.
4. Redirection vanishes exactly for isotropic material and grows with anisotropy. It is unaffected while the minimum tow radius is at most 0.7λ; beyond it the best design found falls to about half by $R \approx 1.2\lambda$ and to 1.75× by $R \approx 2.3\lambda$ (best of 8 starts per limit, a lower bound).
5. A bounded-anisotropy cloak reduces the far-field mismatch 36× at its design frequency, but only 7.5–14.9× at ±5%; under the curvature limit it holds a moderate 2.9–9.6× over $\omega \in [40, 60]$. Transformation directions alone give 1.9×.
6. Band-averaged design delivers multifunctional fiber layouts whose performance holds across the operating band.

Data and code availability. The solver, result files and the scripts that generate the reported results, tables and figures (Appendix F and the parameter tables) are in the repository https://github.com/jtempelman95/wavetopo (branch `paper-revision`; see `docs/paper/README_REPRODUCE.md`).

Competing interests. The author declares no competing interests.

A Complete numerical parameters

Every constant below is the literal argument of the drivers (`examples/dolfinx/dolfinx_lens.py`, `dolfinx_cloak_conforming_vec.py`, `dolfinx_panel.py`) and the analysis scripts in `examples/analysis/`.

Table 7: Complete parameter set.

quantity	lens	cloak
lamina $(E_{f1}, E_{f2}, G_{12}, \nu_{12})$, ρ	$(131, 9, 5, 0.27)$, 1.6	
frequency ω (design)	40	50
structural damping η	0.02	0.02
domain	4×3	8×5, void $R_V = 0.55$ at $(3.2, 2.5)$
mesh	structured, 130×98 ($\times 2$ triangles)	`gmsh` conforming, $h = 0.045$ ($h/2$ on rim)
sponge (c_0, m), Eq. (14)	$(80, 0.6)$	$(70, 0.6)$
source (A, β, x_0), Eq. (15)	$(30, 80, 0.1)$	
CS-RBF (h_s, r)	$(0.33, 0.7)$, mirror-symmetric	$(0.28, 0.6)$, shell $R_{\mathrm{des}} = 1.5$ (tapered)
objective region	box $(0.78L_x, L_y/2) \pm 0.18$	interior, $r > R_C = 1.7$
MMA move	0.10, 70 iterations	0.15 adaptive, 200–240 iterations
curvature limit ζ_{all}	none / front (Table 4)	none / $1/R_V = 1.82$
toolpath pitch d		0.16

B Algorithm

Algorithm 1 Orientation design for wave control (`dolfinx`)

Require: mesh, material $\mathbf{Q}_f$, frequency ω, damping η, sponge $c(\mathbf{x})$, drive b, support sites $\{\mathbf{s}_i\}$, radius r, curvature limit ζ_{all}, void set $\mathcal{V}$

mesh: cut each void in $\mathcal{V}$ from the geometry (`gmsh`/OCC), refining the rim, so $\mathcal{T}_h$ conforms to it and the hole boundary is traction-free ▷ no soft void, no staircase

build $\mathbf{B}, \mathbf{B}_x, \mathbf{B}_y$ from (20), (23); build $\mathbf{S}$ from $\mathbf{S} = \frac{1}{2}(\mathbf{I} - \mathbf{\Pi})$

$\hat{\theta} \leftarrow \mathbf{0}$ ▷ straight-fiber baseline

solve $\mathsf{U}_{\mathrm{ref}}$ on a *separate void-free* mesh and transfer it by non-matching interpolation (cloak only); record the baseline metric

for $\mu \in \{\mu_0, \dots, \mu_{\max}\}$ **do** ▷ penalty continuation

  for $t = 1, \dots, T$ **do**

    $\hat{\theta} \leftarrow \mathbf{S}\hat{\theta}$; $\theta \leftarrow \mathbf{B}\hat{\theta}$ ▷ symmetrize, then interpolate

    assemble $\mathcal{A}(\boldsymbol{\theta})$ as the real 4×4 block (11) ▷ voids already absent from $\mathcal{T}_h$

    forward: factor $\mathcal{A} = \mathbf{LU}$, solve $\mathcal{A}\mathsf{U} = \mathsf{b}$

    evaluate J, and E (focus energy) or the far-field mismatch

    adjoint: solve $\mathcal{A}^{\mathsf{T}}\mathbf{\Lambda} = -\partial J/\partial \mathsf{U}$ (sparse LU of $\mathcal{A}^{\mathsf{T}}$)

    $d_\ell \leftarrow \partial R/\partial\theta_\ell$ by UFL differentiation (19)

    $\boldsymbol{\zeta} \leftarrow \cos\theta\,(\mathbf{B}_x\hat{\theta}) + \sin\theta\,(\mathbf{B}_y\hat{\theta})$; $\mathcal{C} \leftarrow \langle \max(0, \zeta^2 - \zeta_{\mathrm{all}}^2) \rangle$

    $\mathbf{g} \leftarrow$ (19) with the curvature penalty of §5.4 ▷ $\mathbf{S}^{\mathsf{T}}\mathbf{B}^{\mathsf{T}}$ plus curvature term

    $\hat{\theta} \leftarrow \mathrm{MMA}\big(\hat{\theta}, \mathbf{g};\ -\frac{\pi}{2} \le \hat{\theta} \le \frac{\pi}{2}\big)$

  end for

  if $\max_c |\zeta_c| \le \zeta_{\mathrm{all}}$ **then break** ▷ feasible

  end if

end for

delineate toolpaths: sign-consistent branch of $(\cos 2\theta, \sin 2\theta)$; solve $\nabla\Phi = \mathbf{g}$ by `lsqr` (21); tows $\chi = \frac{1}{2} + \frac{1}{2}\cos(2\pi\Phi/d)$, masked off $\mathcal{V}$

return $\hat{\theta}$, θ, U, gain/reduction, χ

C Detailed derivations (modeling)

This appendix carries every modeling step in full, so each equation of §2–§4.3 can be re-derived and independently checked.

C.1 Orthotropic plane-stress stiffness $\mathbf{Q}_f$

In the fiber (material) frame $(1,2)$, under plane stress ($\sigma_{33} = \sigma_{13} = \sigma_{23} = 0$), the orthotropic compliance relating the engineering strain $\boldsymbol{\varepsilon} = [\varepsilon_{11}, \varepsilon_{22}, \gamma_{12}]^\mathsf{T}$ to the stress $\boldsymbol{\sigma} = [\sigma_{11}, \sigma_{22}, \sigma_{12}]^\mathsf{T}$ is

$$\boldsymbol{\varepsilon} = \mathbf{S}\,\boldsymbol{\sigma}, \qquad \mathbf{S} = \begin{bmatrix} 1/E_{f1} & -\nu_{21}/E_{f2} & 0 \\ -\nu_{12}/E_{f1} & 1/E_{f2} & 0 \\ 0 & 0 & 1/G_{12} \end{bmatrix}, \tag{25}$$

with the reciprocity $\nu_{12}/E_{f1} = \nu_{21}/E_{f2}$, i.e. $\nu_{21} = \nu_{12}E_{f2}/E_{f1}$, which makes $\mathbf{S}$ symmetric. The stiffness is $\mathbf{Q}_f = \mathbf{S}^{-1}$. The upper 2×2 block inverts with determinant $\Delta = \frac{1}{E_{f1}E_{f2}}(1 - \nu_{12}\nu_{21})$, giving

$$Q_{11} = \frac{E_{f1}}{1 - \nu_{12}\nu_{21}}, \quad Q_{22} = \frac{E_{f2}}{1 - \nu_{12}\nu_{21}}, \quad Q_{12} = \frac{\nu_{12}E_{f2}}{1 - \nu_{12}\nu_{21}}, \quad Q_{66} = G_{12}, \tag{26}$$

the entries of (2) (`Qf_matrix`). The shear row decouples, hence $Q_{16} = Q_{26} = 0$.

C.2 Rotation to the global frame and $\partial \mathbf{D}_f / \partial \theta$

Let θ be the fiber angle and $(c, s) = (\cos\theta, \sin\theta)$. Rotating the material-frame engineering strain to the global frame is $\boldsymbol{\varepsilon}_{\text{mat}} = \mathbf{T}(\theta)^\mathsf{T}\boldsymbol{\varepsilon}$ with the engineering-strain rotation

$$\mathbf{T}(\theta) = \begin{bmatrix} c^2 & s^2 & -2sc \\ s^2 & c^2 & 2sc \\ sc & -sc & c^2 - s^2 \end{bmatrix}, \tag{27}$$

so that the global stiffness is a congruence transform of the material one,

$$\mathbf{D}_f(\theta) = \mathbf{T}(\theta)\,\mathbf{Q}_f\,\mathbf{T}(\theta)^\mathsf{T}. \tag{28}$$

Differentiating entry-by-entry ($c' = -s$, $s' = c$),

$$\mathbf{T}'(\theta) = \frac{\mathrm{d}\mathbf{T}}{\mathrm{d}\theta} = \begin{bmatrix} -2sc & 2sc & -2(c^2 - s^2) \\ 2sc & -2sc & 2(c^2 - s^2) \\ c^2 - s^2 & -(c^2 - s^2) & -4sc \end{bmatrix}, \tag{29}$$

and by the product rule on (28),

$$\frac{\partial \mathbf{D}_f}{\partial \theta} = \mathbf{T}'\mathbf{Q}_f\mathbf{T}^\mathsf{T} + \mathbf{T}\mathbf{Q}_f\mathbf{T}'^\mathsf{T}, \tag{30}$$

which is the object every orientation sensitivity is built from (cf. (19)).

C.3 Exact five-term Fourier decomposition of $\mathbf{D}_f(\theta)$

The `dolfinx` implementation behind every result in this paper evaluates (28)–(30) directly, as a congruence product differentiated symbolically by UFL. The rotation nevertheless has a sharper structure, which we derive here because it exposes the material invariants through which every orientation effect acts (Sec. 3) and because it gives an equivalent closed form, cheaper to evaluate, that a structured-grid implementation can use: $\mathbf{D}_f(\theta)$ *is a trigonometric polynomial of degree four in* θ, *exactly*.

Start from the entries of $\mathbf{T}(\theta)$. Each is a quadratic monomial in (c, s) — one of c^2, s^2, or sc — and the double-angle identities write every such monomial in the fixed basis $\{1, \cos 2\theta, \sin 2\theta\}$:

$$c^2 = \tfrac{1}{2}(1 + \cos 2\theta), \qquad s^2 = \tfrac{1}{2}(1 - \cos 2\theta), \qquad sc = \tfrac{1}{2}\sin 2\theta. \tag{31}$$

So every entry of $\mathbf{T}$ — and hence of $\mathbf{T}^{\mathsf{T}}$ — lives in $\operatorname{span}\{1, \cos 2\theta, \sin 2\theta\}$. The congruence (28) multiplies two $\mathbf{T}$-factors (with the constant $\mathbf{Q}_f$ between them), so each entry of $\mathbf{D}_f$ is a sum of *products of two* such first-harmonic terms. The product-to-sum identities collapse those products to the second harmonic and no higher,

$$\cos 2\theta \cos 2\theta = \tfrac{1}{2}(1 + \cos 4\theta), \quad \sin 2\theta \sin 2\theta = \tfrac{1}{2}(1 - \cos 4\theta), \quad \cos 2\theta \sin 2\theta = \tfrac{1}{2}\sin 4\theta, \tag{32}$$

so nothing beyond 4θ can appear. Every entry of $\mathbf{D}_f(\theta)$ therefore lies in the five-dimensional space $\operatorname{span}\{1, \cos 2\theta, \sin 2\theta, \cos 4\theta, \sin 4\theta\}$, and collecting the coefficient of each basis function into its own constant 3×3 matrix gives the *exact* decomposition

$$\boxed{\mathbf{D}_f(\theta) = \sum_{s=0}^{4} c_s(\theta)\, \mathbf{C}_s}, \qquad \mathbf{c}(\theta) = \big[\,1,\ \cos 2\theta,\ \sin 2\theta,\ \cos 4\theta,\ \sin 4\theta\,\big], \tag{33}$$

in which the five $\mathbf{C}_s$ depend only on $\mathbf{Q}_f$ (each symmetric, since $\mathbf{D}_f$ is). They are recovered once, offline, by projecting $\mathbf{D}_f(\theta)$ onto the basis $\mathbf{c}$ over a dense θ-sample — an exact fit, not an approximation, because the five functions span the range; in the implementation the residual of the least-squares recovery is 8.5×10^{-16}, i.e. machine zero, which is the numerical confirmation that the degree is really four and not higher.

Two consequences make this the preferred form. First, the derivative no longer needs the product rule (30); it is just the (elementwise) derivative of the scalar basis,

$$\frac{\partial \mathbf{D}_f}{\partial \theta} = \sum_{s=0}^{4} c_s'(\theta)\, \mathbf{C}_s, \qquad \mathbf{c}'(\theta) = \big[\,0,\ -2\sin 2\theta,\ 2\cos 2\theta,\ -4\sin 4\theta,\ 4\cos 4\theta\,\big], \tag{34}$$

which is exact to round-off and shares the constant $\mathbf{C}_s$ with the value (33), so value and gradient cost one small matrix combination each. Second, the decomposition passes *through* the element integral. The fiber element stiffness is $\mathbf{k}_f(\theta) = \int_e \mathbf{B}^{\mathsf{T}} \mathbf{D}_f(\theta)\, \mathbf{B}$, and because the $c_s(\theta)$ are scalars they factor out of the integral,

$$\mathbf{k}_f(\theta) = \sum_{s=0}^{4} c_s(\theta) \underbrace{\int_e \mathbf{B}^{\mathsf{T}} \mathbf{C}_s\, \mathbf{B}}_{\mathbf{k}_f^{(s)}} = \sum_{s=0}^{4} c_s(\theta)\, \mathbf{k}_f^{(s)}, \tag{35}$$

so an implementation can cache five fixed matrices $\mathbf{k}_f^{(s)}$ per element (for a linear triangle, $\mathbf{k}_f^{(s)} = A\, \mathbf{B}^{\mathsf{T}} \mathbf{C}_s \mathbf{B}$, since $\mathbf{B}$ is constant), and every value or θ-derivative of the element stiffness is a five-term weighted sum of them. The two forms give identical stiffness and gradients; the symbolic form is what we use.

C.4 Isotropy annihilation (proof of Remark 1)

$\mathbf{Q}_f$ is isotropic iff $Q_{11} = Q_{22}$ and $Q_{66} = \frac{1}{2}(Q_{11} - Q_{12})$. Then $\mathbf{Q}_f = \lambda \mathbf{1}\mathbf{1}^\mathsf{T} + 2\mu\,\mathbf{P}$ with $\mathbf{1} = [1,1,0]^\mathsf{T}$ and $\mathbf{P}$ the deviatoric projector, both invariant under the in-plane rotation $\mathbf{T}(\theta)$ (an isotropic tensor is unchanged by rotation): $\mathbf{T}\mathbf{Q}_f\mathbf{T}^\mathsf{T} = \mathbf{Q}_f$ for all θ. Hence $\mathbf{D}_f(\theta) \equiv \mathbf{Q}_f$ and, by (30), $\partial\mathbf{D}_f/\partial\theta \equiv \mathbf{0}$; by (19) every orientation sensitivity vanishes. Orientation is a dead variable exactly when the material is isotropic. □

C.5 Elastodynamics, time-harmonic reduction, damping, sponge

Momentum balance for the displacement $\mathbf{u}(\mathbf{x}, t)$ with mass density ρ and the plane-stress constitutive law $\boldsymbol{\sigma} = \mathbf{D}_f(\theta)\boldsymbol{\varepsilon}(\mathbf{u})$ is $\nabla\cdot\boldsymbol{\sigma} = \rho\,\ddot{\mathbf{u}}$. Seeking time-harmonic motion $\mathbf{u}(\mathbf{x}, t) = \mathrm{Re}\{\mathbf{u}(\mathbf{x})e^{+\mathrm{i}\omega t}\}$ turns $\ddot{\mathbf{u}} \to -\omega^2\mathbf{u}$ and $\dot{\mathbf{u}} \to \mathrm{i}\omega\mathbf{u}$. Structural (hysteretic) damping multiplies the stiffness by $(1 + \mathrm{i}\eta)$, and a mass-proportional absorbing sponge $c(\mathbf{x}) \geq 0$ adds a term $+\mathrm{i}\omega c\,\mathbf{u}$ (the $\dot{\mathbf{u}} \to \mathrm{i}\omega\mathbf{u}$ of a viscous force $c\dot{\mathbf{u}}$), giving the strong form (4). Multiplying by a test function $\mathbf{v}$ and integrating by parts (the sponge edges carry the outgoing radiation, the driven edge the source) yields the weak form and, after P_1 interpolation, the complex system (10) with $\mathbf{K} = \int \boldsymbol{\varepsilon}(\cdot)^\mathsf{T}\mathbf{D}_f\boldsymbol{\varepsilon}(\cdot)$, $\mathbf{M} = \int \rho(\cdot)\cdot(\cdot)$, $\mathbf{C} = \int c(\cdot)\cdot(\cdot)$ (`_forms`).

C.6 Exact real four-field split

Write the complex operator of (10) as $\mathbf{D} = (1 + \mathrm{i}\eta)\mathbf{K} - \omega^2\mathbf{M} + \mathrm{i}\omega\mathbf{C} = A + \mathrm{i}B$ with

$$A = \mathbf{K} - \omega^2\mathbf{M}, \qquad B = \eta\mathbf{K} + \omega\mathbf{C}, \tag{36}$$

both real symmetric. Splitting $\mathbf{u} = \mathbf{u}_r + \mathrm{i}\mathbf{u}_i$ and $\mathbf{f} = \mathbf{f}_r + \mathrm{i}\mathbf{f}_i$, the equation $(A + \mathrm{i}B)(\mathbf{u}_r + \mathrm{i}\mathbf{u}_i) = \mathbf{f}_r + \mathrm{i}\mathbf{f}_i$ separates into real and imaginary parts,

$$A\mathbf{u}_r - B\mathbf{u}_i = \mathbf{f}_r, \qquad B\mathbf{u}_r + A\mathbf{u}_i = \mathbf{f}_i, \tag{37}$$

i.e. the real block system (11) on $\mathsf{U} = (u_{xr}, u_{yr}, u_{xi}, u_{yi})$. It is exactly $[\mathrm{Re}, -\mathrm{Im}; \mathrm{Im}, \mathrm{Re}]$ of $\mathbf{D}$, so a real `PETSc` build solves the complex problem without complex scalars.

D Detailed derivations (optimization)

D.1 Discrete adjoint and design sensitivity

Let the assembled forward problem be $\mathcal{A}(\boldsymbol{\theta})\mathsf{U} = \mathsf{b}$ with $\mathcal{A} = [A, -B; B, A]$, and let $J(\mathsf{U})$ be an objective. Form the Lagrangian with a multiplier $\boldsymbol{\Lambda}$, $\mathcal{L} = J(\mathsf{U}) + \boldsymbol{\Lambda}^\mathsf{T}(\mathcal{A}\mathsf{U} - \mathsf{b})$; on the constraint manifold $\mathcal{L} = J$. Its total derivative in a design θ_ℓ is

$$\frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \frac{\partial J}{\partial \mathsf{U}}^\mathsf{T}\frac{\mathrm{d}\mathsf{U}}{\mathrm{d}\theta_\ell} + \boldsymbol{\Lambda}^\mathsf{T}\Big(\mathcal{A}\frac{\mathrm{d}\mathsf{U}}{\mathrm{d}\theta_\ell} + \frac{\partial\mathcal{A}}{\partial\theta_\ell}\mathsf{U}\Big) = \Big(\frac{\partial J}{\partial \mathsf{U}} + \mathcal{A}^\mathsf{T}\boldsymbol{\Lambda}\Big)^\mathsf{T}\frac{\mathrm{d}\mathsf{U}}{\mathrm{d}\theta_\ell} + \boldsymbol{\Lambda}^\mathsf{T}\frac{\partial\mathcal{A}}{\partial\theta_\ell}\mathsf{U}. \tag{38}$$

Choosing $\boldsymbol{\Lambda}$ to annihilate the bracket gives the adjoint equation and reduces the sensitivity to the explicit design dependence of the residual $R(\boldsymbol{\theta}) = \boldsymbol{\Lambda}^\mathsf{T}\big(\mathcal{A}(\boldsymbol{\theta})\mathsf{U} - \mathsf{b}\big)$ (with $\mathsf{U}, \boldsymbol{\Lambda}$ frozen):

$$\mathcal{A}^\mathsf{T}\boldsymbol{\Lambda} = -\frac{\partial J}{\partial \mathsf{U}}, \qquad \frac{\mathrm{d}J}{\mathrm{d}\theta_\ell} = \boldsymbol{\Lambda}^\mathsf{T}\frac{\partial\mathcal{A}}{\partial\theta_\ell}\mathsf{U}. \tag{39}$$

Only $\mathbf{K}(\boldsymbol{\theta})$ (hence A, B) depends on θ, and only through $\partial \mathbf{D}_f / \partial \theta$ (30); expanding $\boldsymbol{\Lambda} = (\boldsymbol{\lambda}_r, \boldsymbol{\lambda}_i)$ block-wise and using $\partial A/\partial \theta_\ell = \partial \mathbf{K}/\partial \theta_\ell$, $\partial B/\partial \theta_\ell = \eta\, \partial \mathbf{K}/\partial \theta_\ell$ gives the explicit cell integral (19). The mass, sponge and source terms carry no θ and drop out, so the gradient is entirely built from $\partial \mathbf{D}_f / \partial \theta$ — which vanishes for an isotropic material, making Remark 1 a statement about the gradient itself. For a quadratic objective $J = \int w|\mathbf{u}|^2$ the adjoint load is linear, $\partial J/\partial \mathsf{U} = 2\,\mathrm{diag}(w)\mathsf{U}$ (a signed weight w gives the multi-region objectives of Sec. 6.6); in the code both the adjoint load and the residual derivative are single `ufl.derivative` calls (`_sensitivity`), FD-verified to $\sim 10^{-6}$.

D.2 CS-RBF chain rule and its analytic derivatives

With $\theta = B\hat{\theta}$ (20), the reduced gradient is $\partial J/\partial \hat{\theta} = B^{\mathsf{T}}(\partial J/\partial \theta)$. For the curvature constraint we also need $\partial_x \theta, \partial_y \theta$. Writing $W_{ci} = \phi(q_{ci})$ and $S_c = \sum_j W_{cj}$ so $B_{ci} = W_{ci}/S_c$, the chain rule on $q_{ci} = \|\mathbf{x}_c - \mathbf{s}_i\|/r$ with $\phi'(q) = -20q(1-q)^3$ gives $\partial_x W_{ci} = \phi'(q_{ci})\,\partial_x q_{ci} = -\frac{20(1-q_{ci})^3}{r^2}(x_c - s_{ix})$, and the quotient rule on B_{ci} yields (23), i.e. $\mathbf{B}_x = S^{-1}W_x - \mathrm{diag}(S^{-1}W_x\mathbf{1})\,B$ (and $\mathbf{B}_y$ likewise); the second term is the derivative of the partition-of-unity normalization, without which $\sum_i (\mathbf{B}_x)_{ci} \neq 0$ and the curvature would be biased (`csrbf_operators`).

D.3 Fiber-curvature functional and its Jacobian

The fiber-path curvature is the directional derivative of θ along the fiber $\mathbf{n} = (\cos\theta, \sin\theta)$, $\zeta = \cos\theta\,\theta_x + \sin\theta\,\theta_y$ with $\theta_x = \mathbf{B}_x\hat{\theta}$, $\theta_y = \mathbf{B}_y\hat{\theta}$. Differentiating w.r.t. $\hat{\theta}$ (product rule, $\partial\theta/\partial\hat{\theta} = B$),

$$\frac{\partial \zeta}{\partial \hat{\theta}} = \mathrm{diag}(-\sin\theta\,\theta_x + \cos\theta\,\theta_y)\,B + \mathrm{diag}(\cos\theta)\,\mathbf{B}_x + \mathrm{diag}(\sin\theta)\,\mathbf{B}_y. \tag{40}$$

The penalty $\mathcal{C} = \langle \max(0, \zeta^2 - \zeta_{\mathrm{all}}^2)\rangle$ has gradient $\partial\mathcal{C}/\partial\hat{\theta} = \frac{2}{n}(\partial\zeta/\partial\hat{\theta})^{\mathsf{T}}[\zeta \odot \mathbf{1}_{\{\zeta^2 > \zeta_{\mathrm{all}}^2\}}]$ (restricted to solid cells for the cloak), FD-verified to $\sim 10^{-10}$ (`curl_penalty`). It is added to the objective with a continuation weight μ ramped until the constraint is met.

D.4 Phase-field toolpath extraction

The tow field is the level set of Φ with $\nabla\Phi = \mathbf{g} = [-\sin\theta, \cos\theta]^{\mathsf{T}}$ (unit, orthogonal to the fiber). This overdetermined system (two equations, one unknown) is solved in the least-squares sense $\min_\Phi \int \|\nabla\Phi - \mathbf{g}\|^2$. Its Euler–Lagrange condition is the Poisson equation $\nabla^2\Phi = \nabla\cdot\mathbf{g}$ with the *natural* Neumann data $\partial\Phi/\partial n = \mathbf{g}\cdot\mathbf{n}$ on $\partial\Omega$. The boundary flux is essential: dropping it (homogeneous Neumann) leaves only $\nabla^2\Phi = \nabla\cdot\mathbf{g}$, whose solution for a nearly divergence-free $\mathbf{g}$ collapses to $\Phi \approx$ const ($|\nabla\Phi| \to 0$), i.e. washed-out tows. We therefore assemble the discrete forward-difference gradient operator $G = [G_x; G_y]$ on the grid and solve $\min_\Phi \|G\Phi - \mathbf{g}\|^2$ directly (`lsqr`, one pinned node fixing the constant), which enforces $|\nabla\Phi| \approx 1$, and hence a constant tow pitch, where the director field is nearly divergence-free. The tows are the peaks of $\chi = \frac{1}{2} + \frac{1}{2}\cos(2\pi\Phi/d)$ (`fiber_phase`). A seam-free director branch is obtained beforehand by region-growing sign propagation (`_orient_consistent`).

D.5 Symmetry projector

The map $\boldsymbol{\Pi}$ exchanging each support site with its mirror image is an involutive permutation ($\boldsymbol{\Pi}^2 = \mathbf{I}$, $\boldsymbol{\Pi}^{\mathsf{T}} = \boldsymbol{\Pi}$), so with $\mathbf{S} = \frac{1}{2}(\mathbf{I} - \boldsymbol{\Pi})$, $\mathbf{S}^{\mathsf{T}} = \mathbf{S}$ and $\mathbf{S}^2 = \frac{1}{4}(\mathbf{I} - 2\boldsymbol{\Pi} + \boldsymbol{\Pi}^2) = \frac{1}{2}(\mathbf{I} - \boldsymbol{\Pi}) = \mathbf{S}$. Thus $\mathbf{S}$ is the *orthogonal* projector onto $\{\hat{\theta} : \boldsymbol{\Pi}\hat{\theta} = -\hat{\theta}\}$, the mirror-symmetric fiber fields. Being self-adjoint,

the same operator symmetrizes the design and the gradient, so imposing symmetry introduces no inconsistency between the two — the projected gradient is exactly the gradient of the restricted problem. □

D.6 MMA subproblem

Each iteration the Method of Moving Asymptotes [35] replaces $J, \mathcal{C}$ by convex separable approximations built from moving asymptotes $L_\ell < \hat{\theta}_\ell < U_\ell$,

$$\tilde{J}(\hat{\theta}) = r + \sum_\ell \Big(\frac{p_\ell}{U_\ell - \hat{\theta}_\ell} + \frac{q_\ell}{\hat{\theta}_\ell - L_\ell} \Big), \quad \begin{cases} p_\ell = (U_\ell - \hat{\theta}_\ell^{(k)})^2 \, (\partial_\ell J)^+ \\ q_\ell = (\hat{\theta}_\ell^{(k)} - L_\ell)^2 \, (\partial_\ell J)^- \end{cases} \tag{41}$$

($(\cdot)^\pm$ the positive/negative parts), with the asymptotes contracted or expanded by the sign history of $\hat{\theta}_\ell$ and a move limit per step. The separable convex subproblem is solved by its (cheap) dual. Because the raw objectives are $O(10^{-4})$ we feed MMA the scale-invariant log-gradient $g_\ell = (\partial_{\theta_\ell} J)/J$ for the lens focus, restoring $O(1)$ sensitivities; the cloak uses the raw gradient (well scaled). Bounds are $\hat{\theta}_\ell \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ (`cfrp_optimizer.py : MMA`).

E Multifunctional panel objective

With the region energies $E_R = \int_{\Omega_R} |\mathbf{u}|^2 \, \mathrm{d}\mathbf{x}$, the four wave jobs compose into a single scalar

$$\mathcal{J}_{\text{wave}} = W_S \log \frac{E_S}{E_S^0} - \lambda_P \frac{E_P}{E_P^0} - \lambda_F \frac{E_F}{E_F^0} - \lambda_O \frac{E_O}{E_O^0}, \tag{42}$$

with the regions made *disjoint* by priority (rims > payload > sensor > outside) so no cell is rewarded and penalized at once, and each term divided by its straight-fiber baseline E_R^0 so all four are $O(1)$ and directly comparable. Two choices matter. The delivery reward is *logarithmic*: with a plain ratio a bright sensor term would swamp the rim penalty and the optimizer would buy delivery by letting the fasteners get busier, whereas log has diminishing returns so protecting the rims stays worthwhile however bright the port already is. The confinement term penalizes interior energy *outside* the target regions, which forces a clean routed beam rather than a diffraction spray. The gradient still needs one adjoint solve per state: for $\mathcal{J}_{\text{wave}} = \sum_R c_R E_R$ (the log term contributing $c_S = W_S/E_S$, re-evaluated each iterate) the adjoint load is the summed region weight $w = \sum_R (\partial \mathcal{J}/\partial E_R) \, \mathbf{1}_{\Omega_R}$, and the discrete-adjoint sensitivity of §5 returns $\mathrm{d}\mathcal{J}_{\text{wave}}/\mathrm{d}\theta_\ell$; the CS-RBF transpose then gives the reduced gradient. The panel is mirror-symmetric about $y = L_y/2$, enforced exactly by a symmetric support grid and the antisymmetric projector of §5.3.

Operational bandwidth. A single-frequency optimum is a single-tone trick. We make the design robust across an operational band [44, 56] by averaging the objective over 5 sample frequencies,

$$\mathcal{J} = \frac{1}{K} \sum_{k=1}^{K} \mathcal{J}_{\text{wave}}(\omega_k; \theta), \qquad \frac{\mathrm{d}\mathcal{J}}{\mathrm{d}\theta_\ell} = \frac{1}{K} \sum_k \frac{\mathrm{d}\mathcal{J}_{\text{wave}}(\omega_k)}{\mathrm{d}\theta_\ell}, \tag{43}$$

one forward and one adjoint solve per frequency per iterate. The band optimization earns its keep at the band *edges*: a design tuned at the center alone holds the shield and delivery across the band but lets the fastener protection degrade toward the edges, which the averaged design restores (Figure 7d).

Table 8: Parameters of the multifunctional panel. Band design: `dolfinx_panel.py --band 44,47,50,53,56 --lam-p 3 --stiff 0.3`; the single-frequency design uses `--band 50` with the same settings.

plate, lamina, damping	7×4; lamina of Table 7; $\eta = 0.02$
mesh	`gmsh` conforming, $h = 0.055$ ($h/2$ on hole rims)
sponge (c_0, m)	$(80,\ 0.6)$ on the right, top and bottom edges
drive	$55\, e^{-80(x-0.15)^2} e^{-((y-2)/0.95)^2}$ (x-directed beam)
fastener holes	radius 0.20 at $(2.55, 1.05)$, $(2.55, 2.95)$, $(4.35, 1.05)$, $(4.35, 2.95)$
rim regions F	annuli of width 0.18 around the holes
payload P	square $(3.2, 2.0) \pm 0.5$
sensor S	square $(5.9, 2.0) \pm 0.3$
weights $(W_S, \lambda_P, \lambda_F, \lambda_O)$	(1, 3, 1, 0.5) (strong shield)
stiffness term	clamped at the four bolt rims, unit downward load over P; $w_c = 0.3$
band	$\omega \in \{44, 47, 50, 53, 56\}$ (single-frequency design: $\omega = 50$)
tow limit	$R = 0.16$ ($\zeta_{\text{all}} = 6.25$); achieved $\max \lvert\zeta\rvert = 5.4$
optimizer	MMA, 110 iterations from straight fibers
CS-RBF	mirror-symmetric support grid, as for the lens

Stiffness. The same orientation is scored for static stiffness. The panel is clamped at the four bolt rims (the bolted mount) and carries a service load over the payload footprint; the compliance $c = \mathbf{f}^\top \mathbf{u}$ and its self-adjoint orientation sensitivity $\mathrm{d}c/\mathrm{d}\theta_\ell = -\int_{\Omega_\ell} \boldsymbol{\varepsilon}(\mathbf{u})^\top (\partial \mathbf{D}_f/\partial\theta)\boldsymbol{\varepsilon}(\mathbf{u})\,\mathrm{d}\mathbf{x}$ — one extra linear solve, reusing the static factorization — enter the objective as $-w_c\, c/c_0$. The fiber angle is largely spent on the wave, so with this weight ($w_c = 0.3$, an explicit term of the objective) the band-optimized layout reaches $1.18\times$ the straight-fiber stiffness at no added mass; pushing stiffness harder trades directly against the routing and shielding. A density/topology field would decouple stiffness and mass from the wave, but a SIMP soft-void contradicts the conforming-void modeling used throughout and is not pursued here.

F Numerical verification

All checks are produced by the scripts `wt_verify.py`, `cloak_fine.py`, `lens_absorber.py` and `lens_flux_control.py` in `examples/analysis/`; Table 9 lists the lens checks.

- **Mesh.** The lens is re-solved with the optimized field resampled (through $(\cos 2\theta, \sin 2\theta)$, the field being a smooth CS-RBF field) on structured meshes with $n_x = 130$ (the design mesh: about 9 P1 elements per wavelength of the slowest, quasi-shear wave at $\omega = 40$, and about 46 per along-fiber qP wavelength), 195 and 260. At $\omega = 40$, G stays within 73–78 and G_φ within 5.72–5.96. The cloak, refined from $h = 0.045$ to 0.03, is discussed in Sec. 6.5.
- **Damping.** At $\omega = 40$ the redirection is independent of the damping ($G_\varphi = 5.8, 5.8, 5.8$ for $\eta = 0.01, 0.02, 0.04$), while the interior growth falls roughly as $1/\eta$ (23.3, 13.4, 6.6: a ratio of 1.7–2.0 per doubling of η), and so does the gain (135, 78, 38). This directly confirms that the gain beyond G_φ is resonant.
- **Absorbing layer.** The reflection coefficient is fitted on the centerline of the straight-fiber plate, $u_x = Ae^{-ikx} + Be^{ikx}$ with the complex along-fiber qP wavenumber, over $0.5 < x < 3.3$ (fit residual $\leq 2\%$); $\lvert R\rvert = \lvert B/A\rvert$ at the sponge entrance. The paper's sponge (width 0.6) gives $\lvert R\rvert = 0.31$–0.34 over $\omega \in [35, 50]$ (Table 9 lists 37.5–50); doubling its width outward, so that the interior is unchanged, gives $\lvert R\rvert \leq 0.13$. Table 9 lists the lens metrics for both.

- **Re-optimization with the better absorber.** The lens is re-optimized from straight fibers with the doubled sponge (same objective, MMA settings, 70 iterations and mirror-symmetric CS-RBF field; `lens_absorber.py`). At $\omega = 40$ it reaches $G = 70\times = 4.4\times$ (G_φ) $\times\, 15.9\times$ (growth), with $G_\varphi \geq 3.6\times$ over $[40, 60]$ and a band average of $3.9\times$, equal to that of the original design evaluated with the same absorber ($3.9\times$). The original design does not exploit the reflections; the reflections raise the redirection metric of any design.
- **Panel mesh.** The band design re-evaluated on a finer conforming mesh ($h = 0.04$ instead of 0.055; `panel_mesh_check.py`) keeps every qualitative result across the band: the sensor at or above $1.08\times$, the payload at or below $0.27\times$ and the rims at or below $0.53\times$ their straight-fiber energies. Sensor and payload ratios change by at most 6%; the rim ratios, the smallest energies, change most (by up to 0.10 in absolute terms).
- **Cloak frequency response on the finer mesh.** The cloak re-optimized on $h = 0.03$ gives $7.4\times$ and $14.6\times$ at $\omega = 47.5$ and 52.5 against $40\times$ at 50 (`cloak_fine_band.py`), as narrowband as the design on the coarser mesh.

Table 9: Numerical verification of the lens (fixed optimized design).

check	setting	G	G_φ	growth
lens mesh, $\omega = 40$	$n_x = 130$	78.0	5.82	13.41
lens mesh, $\omega = 40$	$n_x = 195$	73.2	5.72	12.80
lens mesh, $\omega = 40$	$n_x = 260$	74.4	5.96	12.48
lens mesh, $\omega = 50$	$n_x = 130$	5.5	6.01	0.91
lens mesh, $\omega = 50$	$n_x = 195$	5.4	5.95	0.90
lens mesh, $\omega = 50$	$n_x = 260$	5.7	6.32	0.90
lens damping, $\omega = 40$	$\eta = 0.01$	135.1	5.80	23.32
lens damping, $\omega = 50$	$\eta = 0.01$	5.1	6.03	0.85
lens damping, $\omega = 40$	$\eta = 0.02$	78.0	5.82	13.41
lens damping, $\omega = 50$	$\eta = 0.02$	5.5	6.01	0.91
lens damping, $\omega = 40$	$\eta = 0.04$	37.9	5.77	6.57
lens damping, $\omega = 50$	$\eta = 0.04$	5.9	5.94	1.00
lens sponge, $\omega = 37.5$	width 0.6, $\lvert R\rvert = 0.31$	6.6	3.74	1.77
lens sponge, $\omega = 37.5$	width 1.2, $\lvert R\rvert = 0.12$	4.0	3.00	1.32
lens sponge, $\omega = 40$	width 0.6, $\lvert R\rvert = 0.34$	78.0	5.82	13.41
lens sponge, $\omega = 40$	width 1.2, $\lvert R\rvert = 0.11$	64.1	4.38	14.62
lens sponge, $\omega = 45$	width 0.6, $\lvert R\rvert = 0.33$	16.3	6.54	2.49
lens sponge, $\omega = 45$	width 1.2, $\lvert R\rvert = 0.11$	10.3	4.56	2.26
lens sponge, $\omega = 50$	width 0.6, $\lvert R\rvert = 0.32$	5.5	6.01	0.91
lens sponge, $\omega = 50$	width 1.2, $\lvert R\rvert = 0.11$	6.7	4.61	1.46

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